ON THE RELATION BETWEEN TWO ROBUSTNESS FORMULATIONS: ROBUSTNESS IN STABILITY TO PERTURBATIONS (RSP) AND DISTRIBUTIONALLY ROBUST OPTIMIZATION (DRO) *

JOFFRE DECORE, JAN OBŁÓJ AND SERDAR YÜKSEL †

Abstract. In Bayesian decision, stochastic optimization and stochastic control applications, typically only an ideal probability model is assumed and the decision/control design is based on the given model, raising the problem of performance loss due to the mismatch between the assumed model and the actual model. To study the effect of the mismatch, two common formulations on robustness that have been considered are the following: (a) Distributionally Robust Optimization (DRO) framework, where a model is taken to be among a neighborhood of models around a nominal one and the design is made as a minimax optimizer among this neighborhood of models, and (b) Robustness in Stability to Perturbations (RSP), a term introduced in this paper but a paradigm which is even more common than DRO, in which case the robustness is with respect to the application of the design for an incorrect model but applied to the true model. The goal is to arrive at a loss which decays to zero as the incorrect model approaches the true model (in a suitable sense; such as weak convergence, continuous weak convergence, adapted Wasserstein, or adapted weak convergence etc.). We establish the following: (i) Asymptotic DRO and RSP are generally not comparable as they do not imply one other. This is established via two counterexamples, both for static and dynamic problems. (ii) Asymptotic DRO generally implies value convergence. (iii) Conditions which allow for the exchange of min and max in the DRO formulation via Sion's minimax theorem imply that DRO is equivalent to value convergence. (iv) However, for many applications, including single-stage empirical Bayesian decision problems and continuous-convergence of kernels, value congergence and RSP are known to be simultaneously satisfied, and accordingly DRO does imply RSP for such practical applications. (v) As a by-product of our analysis, for the dynamic programming recursions under the DRO formulation, we develop several regularity properties facilitating dynamic programming, measurable selection and the applicability of Sion's minimax theorem. During our analysis, we also re-interpret several results in the literature and show their equivalence with RSP.

AMS subject classifications. 93E20, 93E03, 93E11, 62G35

1. Introduction: Robust Design to Incorrect Models.

Robustness is a fundamental consideration in applied mathematics, engineering, finance and sciences at large: Scientific reasoning based on modeling would not be as successful if our approach, interpretations or designs were not robust to errors in modeling. In particular, in Bayesian decision, stochastic optimization and stochastic control applications, typically only an ideal probability model is assumed and the decision/control design is based on the given model, raising the problem of performance loss due to the mismatch between the assumed model and the actual model.

Two popular paradigms have emerged for studying robustness. One is with regard to stability of control policy performance to small perturbations (which we will refer to as Robustness as Stability to Perturbations (RSP) in this paper), and the other is the Distributionally Robust Optimization Paradigm (to be denoted as DRO). In this paper, we present a comparative analysis between these two paradigms.

DRO Paradigm. Under the distributionally robust optimization (DRO) framework, it is assumed that the underlying probability measure of the system lies within an ambiguity set and a worst case single-stage optimization is made considering the probability measures in the ambiguity set. There have been many different interpretations and constructions on the ambiguity sets. The ambiguity can be on priors, on conditional transition kernels which may depend on the entire history, the current state, or the current time; or the ambiguity may only be with respect to a driving noise. The recent studies [44, 17, 13] provide different perspectives on the mathematical formulation, beyond the particular topologies on ambiguity sets which also have had various flavours: To construct ambiguity sets, [18, 27] use the Wasserstein metric, [26] uses the Prokhorov metric which metrizes the weak topology, [60] has

* This research was initiated during an Isaac Newton Institute Workshop. The author(s) would like to thank the Isaac Newton Institute for Mathematical Sciences, Cambridge, for support and hospitality during the programme [34;Bridging Stochastic Control And Reinforcement Learning: Theories and Applications] where work on this paper was initiated. This work was supported by EPSRC grant no EP/R014604/1.#34

†The authors are with

studied a min-max formulation where the one-stage transition kernel belongs to a ball under the total variation metric for each state action pair (see also [59] which uses the total variation metric under a different formulation), and [42] works with relative entropy. [63, 36, 47] have studied robust dynamic programming approaches through a min-max formulation for fully observed finite state-action space models with uncertain transition probabilities.

Closely related to the DRO approach in its essence, and equivalent to it when the balls are characterized by relative entropy between measures, a further common approach in the literature has been to consider risk sensitive formalism: The connections of this formulation to risk sensitive control were established in [37, 25]. In this context, relative entropy constraints have been used in the literature to probabilistically model the uncertainty class, see e.g. [52, Eqn. (4)] or [25, Eqns. (2)-(3)]. Here, one selects a nominal system which satisfies a relative entropy bound between the actual measure and the nominal measure, solves a risk sensitive optimal control problem under the nominal model, and this solution value provides an upper bound for the original system performance over the entire uncertainty class. As such, a common approach in robust stochastic control has been to consider all models which satisfy certain bounds in terms of relative entropy pseudo-distance (or Kullback-Leibler divergence); see e.g. [25, 52, 51, 19] among others.

Instead of a stochastic formulation, design of controllers that perform sufficiently well for all possible systems within deterministic and structured constraints, such as H_∞ norm bounded perturbations, has been studied extensively in classical control theory [8, 12, 71]. Further related work with model uncertainty includes [48, 16, 65], with some further work in the economics literature [32, 31].

In summary, under the DRO paradigm, the design builds on a game theoretic formulation where the minimizer is the decision maker/optimizer/controller and the maximizer is the uncertainty.

RSP Paradigm via Continuous Convergence of Kernels and Adapted Topologies. In another prominent and perhaps more commonly occurring paradigm, one designs a policy for an approximate model and applies the policy to a true system: The mismatch loss which arises due to the application of the design for the incorrect model but applied to the true model is to be small as the models approach one another. A precise definition will be given later in the paper.

Under continuous convergence of kernels, [41] in discrete-time and [53] in continuous-time for controlled diffusion problems established robustness in the sense of RSP. Closely related to such results and continuous convergence, a topology on spaces of probability measures corresponding to laws of stochastic processes, which has been used in a wide variety of contexts in stochastic analysis, is defined by the following. A sequence of stochastic processes is said to converge to another process if their finite-dimensional marginals converge weakly, and their conditional distributions of future variables given the past (viewed as measure-valued stochastic processes) also converge weakly. Aldous has termed this *extended weak convergence* [1] and Hellwig has named it *the information topology* [33]; these have recently been shown to be equivalent in discrete-time [6, 49]. The *adapted Wasserstein metric* [10, 6, 15] has been shown to possess strong robustness properties in a variety of applications [14, 5, 11], not unlike the continuous weak convergence notion noted above, under which robustness in the sense of RSP is established (see [56] for a comprehensive review).

[MORE Here..Papers by Jan / Johannes / etc.]

We also note that approximation methods for stochastic control problems with standard Borel spaces through quantization, which lead to finite models, can be viewed as approximations of transition kernels, but this interpretation requires caution: indeed, [57, 2, 3], among many others, study approximation methods for Markov decision processes (MDPs) where the convergence of approximate models is satisfied in a particularly constructed fashion. In particular, [57] presents a construction for the approximate models through quantizing the actual model with continuous spaces (leading to a finite space model), which allows for continuity

and robustness results with only a weak continuity assumption on the true transition kernel which, in turn, leads to *continuous weak convergence* of approximate models as discussed in [39] and thus becomes a very important special case of robustness in the sense of RSP. A detailed analysis of approximation methods for continuous state and action spaces can be found in [55] for both fully observed and partially observed models. Under more stringent conditoins on the regularity properties, modulus of continuity under suitable kernel topologies have also been obtained: [21, 72, 56].

Value Convergence and Continuity of Optimal Cost in Models. An intermediary concept of robustness, which has been studied as a separate criterion also in the literature, is the continuous dependence of optimal cost in models, without robustness taken into account. For fully observed models, [43, Theorem 5.1] establishes continuity results for approximate models and gives a set convergence result for sets of optimal control actions, but this set convergence result is inconclusive for robustness without further assumptions on the true system model. Reference [46] is another related work which studies continuity of the value function for fully observed models under a general metric defined as the integral probability metric, which captures both the total variation metric or the Kantorovich metric with different setups (which is not weaker than the metrics leading to weak convergence). Related supper semi-continuity and lower semi-continuity results have been obtained in convergence for measurement kernels in [69].

Robustness to Priors. While in the above, we primarily focused on robustness due to modeling errors in system dynamics, in many applications the uncertainty is with regard to the prior probability models [cite robust statistics paper]. Related work also includes recent studies [69, 38]. [69] considers various topologies on the sets of observation channels in partially observed stochastic control and provides some supporting results. [64] and [40] present robustness and continuity properties with respect to prior measures.

1.1. Contributions.

(i) Asymptotic DRO and RSP are generally not comparable as they do not imply one another. This is established via two counterexamples both for single-stage and multi-stage dynamic problems: Theorem 2.3 shows the lack of implication relationship in the single-stage setup, while Theorem 3.1 establishes the result for the dynamic problem. We note that the result on DRO not implying RSP may be viewed as a surprising one.

(ii) DRO does not generally imply value convergence, but Lemma 2.7 shows that DRO does imply upper semicontinuity of the optimal value at the nominal model. Accordingly, when the optimal value is also lower semicontinuous, DRO implies value convergence, as established in Theorem 2.8.

(iii) We identify conditions for which DRO and RSP can be related through value convergence in single-stage Bayesian decision problems. In particular, under suitable continuity conditions, DRO implies RSP, and when the hypotheses of Sion's minimax theorem hold, DRO and RSP are equivalent. This is shown in Theorem 2.9. Furthermore, we provide sufficient conditions for which DRO, value convergence, and RSP hold simultaneously for standard empirical Bayesian decision problems.

(iv) We extend our comparison of DRO and RSP to infinite-horizon discounted cost dynamic control problems. For appropriately constructed ambiguity classes, we formulate the DRO problem through state-wise dynamic programming recursions and identify when they are well-defined. Under weak convergence, Proposition 3.10 shows that the iterations converge to a fixed point and Theorem 3.12 shows that this fixed point characterizes the DRO value. Furthermore, Theorem 3.13 shows that the DRO value converges to the nominal optimal value as the ambiguity sets shrink.

(v) As a by-product of our analysis, for the dynamic programming recursions under the DRO formulation, we identify conditions with which the minimax theorem is applicable and develop several regularity properties. These, under both weak and

total variation convergence topologies, lead to measurable selection and associated existence and well-posedness results for dynamic programming. These are new contributions to the DRO literature.

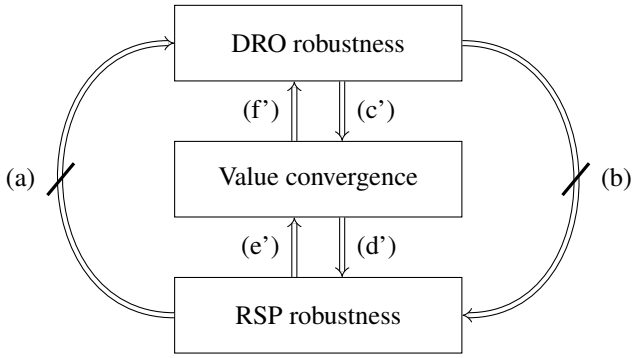


Fig. 1.1: Implication relationships among the robustness formulations studied in this paper. (a) in general, we have that RSP robustness does not imply DRO robustness. (b) in general, we have that DRO robustness does not imply RSP robustness. (c') If $J^*(\cdot, Q)$ is lower semicontinuous at P, then DRO implies value convergence. (d')–(e') If $|J(P_n, Q, \gamma^*_{P_n}) - J(P, Q, \gamma^*_{P_n})| \to 0$ for every $P_n \to P$ weakly, then value convergence and RSP are equivalent. (f') Under a randomized-policy formulation such that Sion's minimax theorem holds, value convergence implies DRO. Respectively, see Theorem 2.8 and Theorem 2.9. (c')–(f') highlight the implication relationships for the static single-stage setup which lead to similar arguments in the dynamic case; see Theorem 3.3 and Theorem 3.4.

Our primary focus will be on multi-stage dynamic problems, however, due to its instructional value in view of relative simplicity and broad applicability, we will start our analysis with an application on robustness (including empirical consistency) of Bayesian decision problems.

2. Static (Single-Stage) Setup: On Robustness and Asymptotic Consistency of Bayesian Decision Making under RSP and DRO Formulations. Let $\mathbb{X}$ be a state space. Let a nominal prior measure be denoted by $P \in \mathcal{P}(\mathbb{X})$, where $\mathcal{P}(\mathbb{X})$ is the set of probability measures on $\mathbb{X}$, a measurement kernel/channel be given with $Q \in \mathcal{Q}$ and the decision/control action be selected under an admissible control policy from the set of such policies Γ: Here, Γ is the set of Borel measurable functions from $\mathbb{Y}$ to $\mathbb{U}$, where $\mathbb{Y}$ and $\mathbb{U}$ are measurement and action spaces respectively. The optimal expected cost is defined as,

$$J^*(P, Q) = \inf_{\gamma \in \Gamma} \int_{\mathbb{X} \times \mathbb{Y}} c(x, \gamma(y)) Q(dy|x) P(dx).$$

See Figure 2.1 for an illustration.

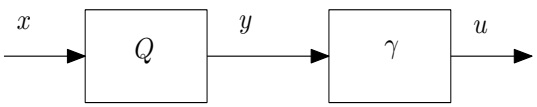


Fig. 2.1: Bayesian Decision Problem Setup

Throughout this section, we will impose the following:

ASSUMPTION 2.1. *The measurable cost function satisfies $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}_+$, and for every $x \in \mathbb{X}$, $c(x, \cdot) : \mathbb{U} \to \mathbb{R}_+$ is continuous. Furthermore $\mathbb{U}$ is compact.*

If $c(x, \cdot) : \mathbb{U} \to \mathbb{R}$ is continuous, by a standard measurable selection criterion (see Appendix D of [34] or [70]), an optimal γ_P^* exists for any given prior P.

DEFINITION 2.1. *Let $P_n \to P$ in some topology (such as weak convergence), and with*

$$\inf_{\gamma \in \Gamma} E_{P_n}[c(X, \gamma(Y))] = E_{P_n}[c(X, \gamma_{P_n}^*(Y))],$$

so that $\gamma_{P_n}^*$ is an optimal decision under the prior P_n. The Bayesian decision problem above is robust in the sense of Stability to Perturbations (RSP) (with the topology noted) if

$$\left| E_P[c(X, \gamma_{P_n}^*(Y))] - E_P[c(X, \gamma_P^*(Y))] \right| \to 0 \tag{2.1}$$

While we do not aim to restrict the notion of convergence $P_n \to P$ to be necessarily induced by a metric (e.g., setwise convergence may also be considered), in applications typically a metric will be implied, in which case the above definition can be refined as *Robust in the sense of Stability to Perturbations (RSP) under ρ*.

Once again, let ρ be a metric on $\mathcal{P}(\mathbb{X})$.

DEFINITION 2.2. *We will say that the Bayesian decision problem above is robust in the sense of (Asymptotic) Distributionally Robust Optimization (DRO) under ρ if*

$$\lim_{n \to \infty} \left| \inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{P}(\mathbb{X}) : \rho(\bar{P}, P) \leq \frac{1}{n}} E_{\bar{P}}[c(X, \gamma(Y))] - E_P[c(X, \gamma_P^*(Y))] \right| = 0 \tag{2.2}$$

Notation. As a quick note on notation, we will define ambiguity balls as follows:

Define for $P \in \mathcal{P}(\mathbb{X})$, $r > 0$,

$$\mathcal{B}_r^\rho(P) := \{\bar{P} \in \mathcal{P}(\mathbb{X}) : \rho(\bar{P}, P) \leq r\}.$$

Therefore, (2.2) is equivalent to

$$\lim_{n \to \infty} \left| \inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}_{1/n}^\rho(P)} E_{\bar{P}}[c(X, \gamma(Y))] - E_P[c(X, \gamma_P^*(Y))] \right| = 0 \tag{2.3}$$

As a final note on notation, we may use $C_b(\mathbb{X})$ to denote the continuous and bounded functions of $\mathbb{X}$, and $B_b(\mathbb{X})$ to denote the measurable and bounded functions of $\mathbb{X}$.

An Application and on the Notions of Convergence. A typical application for the above problem is for the case where a prior measure is learned from $i.i.d.$ data. Let a random variable be repeatedly observed, but its probability measure not be known apriori. Let $\{(X_i),\, i \in \mathbb{N}\}$ be an $\mathbb{X}$-valued i.i.d. random variable sequence generated according to some distribution μ. Defining for every (fixed) Borel $B \subset \mathbb{X}$, and $n \in \mathbb{N}$, the empirical occupation measures

$$\mu_n(B) = \frac{1}{n} \sum_{i=1}^{n} 1_{\{X_i \in B\}},$$

one has $\mu_n(B) \to \mu(B)$ almost surely by the strong law of large numbers. As the above also applies for countable collections of test functions of the form $f(X_i)$, and such a collection can be used to determine weak convergence [28, Theorem 3.4.5], it follows that $\mu_n \to \mu$ weakly with probability one ([24], Theorem 11.4.1). However, μ_n cannot converge to μ in total variation or setwise, in general. On the other hand, if we know that μ admits a density, we can find estimators to estimate μ under total variation [22, Chapter 3]. The same

applies for the Wasserstein convergence under additional conditions on the moments [62, 23, 29]. In the learning theoretic context, the convergence of optimal costs under μ_n to the cost optimal for μ is called the *consistency of empirical risk minimization* (see [61] for an overview). Therefore, weak convergence inducing metrics as well as the Wasserstein metric are operationally significant metrics on probability measures and transition kernels both for (2.1) and (2.2)

2.1. Counterexamples to Equivalence. In the following we provide two counterexamples to highlight the distinction between the two robustness frameworks.

EXAMPLE 2.1. *In the following, we show that RSP does not imply DRO, in general. A reflection on this result will guide our analysis to be presented later.*

Let

$$R_n = (1 - \frac{1}{n})Z_n + \frac{1}{n}S$$

where $S \sim \mathcal{N}(0,1)$ (standard Gaussian with mean 0 and unit variance). Let $Z_n \sim \delta_{\frac{1}{n}}$ and $X_n := (R_n, Z_n) \sim P_n$. Clearly $P_n \to P = \delta_{(0,0)}$ weakly.

Let for $x \in \mathbb{R}^2$, $Q(dy|x) = \mathrm{Unif}([-1,1])$, so that Q is a non-informative channel.

Let the cost criterion be given as:

$$J(P_n, \gamma) = \int r^2 e^{r^2} (z-u)^2 P_n(dr, dz),$$

where $u = \gamma(y)$ as earlier.

In this case, observe that $U_n = \gamma^*_{P_n}(Y) = Z_n = \frac{1}{n}$, almost surely, leads to a cost of 0. That is,

$$\inf_{\gamma \in \Gamma} E_{P_n}[c(X, \gamma(Y)] = E_{P_n}[c(X, \gamma^*_{P_n}(Y)] = 0$$

Furthermore, $J(P, \gamma^*_{P_n}) = 0$ as well since $P = \delta_{(0,0)}$.

Under DRO, however; for all $n \in \mathbb{N}$, we have that

$$\inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} E_{\bar{P}}[c(X, \gamma(Y))] = \infty$$

since $\frac{1}{n} r^2 e^{r^2}$ is not integrable under the standard Gaussian measure for any finite n, and therefore

$$\liminf_{n \to \infty} \inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} E_{\bar{P}}[c(X, \gamma(Y))] = \infty.$$

Thus, RSP holds, while DRO fails.

EXAMPLE 2.2. *In the following we show that DRO does not imply RSP, in general.*

Let $Q(dy|x) = \mathrm{Unif}([-1,1])$, so that Q is a non-informative channel. Let

$$\mathbb{U} = [0,1], \qquad \mathbb{X} = \{0\} \cup \{1/n : n \in \mathbb{N}\}.$$

Let $P = \delta_0$, and for $n \in \mathbb{N}$, $P_n = \delta_{1/n}$. Endow $\mathcal{P}(\mathbb{X})$ with any metric ρ which metrizes weak convergence. Once again, $P_n \to P$ weakly.

Define the cost function $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}_+$ by

$$c(0, u) = u^2$$

and, for $n \in \mathbb{N}$

$$c(1/n, u) = \frac{1}{n}(1-u)^2$$

Note that this cost function is bounded, measurable in x, and continuous in u for every fixed x. Moreover, $\mathbb{U}$ *is compact.*

At the true prior $P = \delta_0$*, we have that*

$$J(P, \gamma) = c(0, u) = u^2,$$

where $u = \gamma(y)$*. Thus, our unique optimal action is* $\gamma_P^*(Y) = 0$*, and* $J^*(P) = 0$*.*

However, at the perturbed prior $P_n = \delta_{1/n}$*,*

$$J(P_n, \gamma) = c(1/n, u) = \frac{1}{n}(1-u)^2.$$

Thus, its unique optimal action is $\gamma_{P_n}^*(Y) = 1$ *and* $J^*(P_n) = 0$*.*

When we apply $\gamma_{P_n}^*$ *to the true model* P*, we get*

$$J(P, \gamma_{P_n}^*) = c(0, 1) = 1.$$

Thus, $J(P, \gamma_{P_n}^*) \not\to J^*(P)$*. That is, RSP fails.*

Now, we must show that DRO holds. For $r > 0$*, define*

$$V(r) := \inf_{u \in [0,1]} \sup_{\bar{P} \in \mathcal{B}_r^\rho(P)} J(\bar{P}, u).$$

Since $J^*(P) = 0$ *and the cost is nonnegative, we have that* $V(r) \geq 0$*. We obtain an upper bound by choosing the admissible action* $\gamma(y) = 0$*. Then*

$$c(0, 0) = 0, \qquad c(1/n, 0) = \frac{1}{n}.$$

Thus, the function $g(x) := c(x, 0)$ *satisfies*

$$g(0) = 0, \qquad g(1/n) = \frac{1}{n}.$$

Therefore, $g \in C_b(\mathbb{X})$*. Since* ρ *metrizes weak convergence, and* $P = \delta_0$*, we claim*

$$\sup_{\bar{P} \in \mathcal{B}_r^\rho(P)} \int g(x) \bar{P}(dx) \to 0 \qquad \text{as } r \downarrow 0.$$

If this were false, there would exist $\epsilon > 0$*, a sequence* $r_m \downarrow 0$*, and measures* $\bar{P}_m$ *with* $\rho(\bar{P}_m, P) \leq r_m$*, such that*

$$\int g(x) \bar{P}_m(dx) \geq \epsilon$$

for all m*. However,* $\rho(\bar{P}_m, P) \leq r_m \to 0$*, so* $\bar{P}_m \to P$*. Since* $g \in C_b(\mathbb{X})$*, weak convergence gives*

$$\int g(x) \bar{P}_m(dx) \to \int g(x) P(dx) = g(0) = 0$$

which is a contradiction.

Consequently,

$$0 \leq V(r) \leq \sup_{\bar{P} \in \mathcal{B}_r^\rho(P)} J(\bar{P}, 0) = \sup_{\bar{P} \in \mathcal{B}_r^\rho(P)} \int g(x) \bar{P}(dx) \to 0.$$

Thus, $V(r) \to 0 = J^*(P)$*. That is,*

$$\inf_{u \in [0,1]} \sup_{\bar{P} \in \mathcal{B}_{1/n}^\rho(P)} J(\bar{P}, u) \to J^*(P).$$

Thus, DRO holds, while RSP fails.

Thus, we have the following.

THEOREM 2.3. *In general,*

(i) Robustness in the sense of RSP (2.1) does not imply robustness in the sense of DRO (2.2), and

(ii) Robustness in the sense of DRO (2.2) does not imply robustness in the sense of RSP (2.1).

Proof. Part (i) follows from Example 2.1, while part (ii) follows from Example 2.2. ◇

REMARK 2.1 (A Natural Conjecture is Incorrect.). *While it may appear to propose as a natural conjecture that DRO should be a strictly more conservative notion than RSP due to its formulation, Example 2.2 highlights that there is no implication relationship in general. DRO concerns policies which solve the minimax problem over an ambiguity set around* P *(and thus may allow for a more restrictive class of policies), while RSP concerns policies which are optimal for each individual perturbed priors* P_n *and thus leads to a more general class of policies: these two classes of policies need not coincide.*

2.2. Conditions for Equivalence via Sion's Minimax Theorem. Having the counterexamples above, we study equivalence conditions between the two formulations.

2.2.1. RSP Robustness to Incorrect Priors. As earlier, consider the single stage stochastic control problem with cost function c and initial distribution P. The DM computes an optimal policy, $\widetilde{\gamma}^*$, for the problem with $\widetilde{P}$ and applies it to the problem with the true prior P.

The first result below shows that $J^*(P, Q)$ is not necessarily continuous in prior measures under weak convergence. We will then present an upper semi-continuity result and some sufficiency continuity conditions for weak convergence in prior measures.

THEOREM 2.4. *[40, Theorem 2.2] Given a channel* $Q \in \mathcal{Q}$*,* $J^*(P, Q)$ *is not necessarily continuous in* P *under weak convergence. This holds even when* $\mathbb{X}$*,* $\mathbb{Y}$*, and* $\mathbb{U}$ *are compact and when* c *is bounded and continuous on* $\mathbb{X} \times \mathbb{U}$*.*

ASSUMPTION 2.2. Q *is continuous in total variation in the sense that as* $x_n \to x$*,* $\|Q(dy|x_n) - Q(dy|x)\|_{TV} \to 0$*.*

THEOREM 2.5. *[40, Theorem 2.5] Let a cost function* $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}$ *be given and let Assumption 2.2 hold. Further, let* $c(x, u)$ *be bounded and continuous on* $\mathbb{X} \times \mathbb{U}$*, and* $\mathbb{U}$ *be compact. If* $P_n \to P$ *weakly, then* $J^*(P_n, Q) \to J^*(P, Q)$*.*

PROPOSITION 2.1. *[40, Proposition 3.2] Let an initial distribution* $P \in \mathcal{P}(\mathbb{X})$*, and a cost function* $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}$ *be given. Assume that* $P_n \to P$ *weakly and let* $\gamma^*_{P_n}$ *be an optimal policy for the control problem* $\Xi_n = (c, P_n)$*. Then, it is not necessarily true that* $J(P, Q, \gamma^*_{P_n}) \to J^*(P, Q)$ *as* $n \to \infty$*. This result holds even if* c *is bounded and continuous in both* x *and* u*.*

Assumption 2.2, however, leads to the following positive result.

THEOREM 2.6. *[40, Theorem 3.1] Suppose that* $c(x, u)$ *is bounded,* $\mathbb{U}$ *is compact and Assumption 2.2 holds. Then, as* $P_n \to P$ *weakly,* $\lim_{n\to\infty} |J(P, Q, \gamma^*_{P_n}) - J^*(P, Q)| \to 0$*, that is the system is robust to errors in the sense of RSP under weak convergence.*

2.2.2. DRO Convergence and Value Convergence. It may appear natural that robustness in the sense of DRO should imply value convergence for every sequence of priors contained in the shrinking ambiguity sets. This, however, does not hold in general.

EXAMPLE 2.3. *Consider the same setup as Example 2.2, however we change the cost function to be defined as*

$$c(0, u) = 1,$$

and, for $n \in \mathbb{N}$*,*

$$c(1/n, u) = 0.$$

Therefore, we have

$$J^*(P, Q) = 1, \qquad \textit{and} \qquad J^*(P_n, Q) = 0.$$

Clearly, $J^*(P_n, Q) \not\to J^*(P, Q)$*, however,*

$$1 = J^*(P, Q) \leq \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q) \leq 1.$$

That is,

$$\inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q) = 1 = J^*(P, Q)$$

The above demonstrates that robustness in the sense of DRO only provides an upper bound on the perturbed optimal values and does not, in general, imply value convergence.

LEMMA 2.7. *Robustness in the sense of DRO (2.2) implies* $J^*(\cdot, Q)$ *is upper semicontinuous at* P*.*

Proof. Observe that

$$\begin{aligned}
&\inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} E_{\bar{P}}[c(X, \gamma(Y))] \\
&\geq \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} \inf_{\gamma \in \Gamma} E_{\bar{P}}[c(X, \gamma(Y))] \\
&= \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} E_{\bar{P}}[c(X, \gamma^*_{\bar{P}}(Y)]
\end{aligned}$$

That is, if DRO holds, we have

$$\sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} J^*(\bar{P}, Q) \to J^*(P, Q).$$

Suppose $P_n \to P$ under ρ. For every $k \in \mathbb{N}$, there exists N such that, for all $n \geq N$, $P_n \in \mathcal{B}^{\rho}_{1/k}(P)$. Therefore,

$$\limsup_{n \to \infty} J^*(P_n, Q) \leq \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/k}(P)} J^*(\bar{P}, Q).$$

Taking $k \to \infty$ gives

$$\limsup_{n \to \infty} J^*(P_n, Q) \leq J^*(P, Q).$$

Thus, $J^*(\cdot, Q)$ is upper semicontinuous at P. $\diamond$

THEOREM 2.8. *Let ρ metrize weak convergence, and suppose that $J^*(\cdot, Q)$ is lower semicontinuous at P. Then, robustness in the sense of DRO (2.2) implies value convergence. That is, when (2.2) holds,*

$$J^*(P_n, Q) \to J^*(P, Q)$$

for every $P_n \in \mathcal{B}^{\rho}_{1/n}(P)$.

Proof. Suppose DRO holds. Let $P_n \in \mathcal{B}^{\rho}_{1/n}(P)$. As $P_n \to P$ weakly under ρ, and by the lower semicontinuity of $P \mapsto J^*(P, Q)$ at P and by Lemma 2.7, we have that

$$J^*(P, Q) \leq \liminf_{n\to\infty} J^*(P_n, Q) \leq \limsup_{n\to\infty} J^*(P_n, Q) \leq J^*(P, Q).$$

Thus, this gives the value convergence we require. $\diamond$

Given this setup and relationship, DRO implies RSP when value convergence is equivalent to RSP [40].

2.2.3. Equivalence of RSP and DRO convergence via Sion's Minimax Theorem. If we can swap the order of the $\inf$ and $\sup$

$$\left| \inf_{\gamma\in\Gamma} \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} E_{\bar{P}}[c(X, \gamma(Y))] - E_P[c(X, \gamma^*(Y))] \right|$$

to arrive at

$$\left| \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} \inf_{\gamma\in\Gamma} E_{\bar{P}}[c(X, \gamma(Y))] - E_P[c(X, \gamma^*(Y))] \right|$$

Then, the above indeed becomes an instance of the continuity of value. A sufficient condition is via Sion's minimax theorem, by endowing policies with a convex compact formulation under the Young topology under the condition that $Q(dy|x) = f(x, y)\psi(dy)$ for some reference measure ψ and f being the Radon-Nikodym derivative. See [20] or [66, Section 3.1].

THEOREM 2.9. *Let ρ metrize weak convergence. Suppose that $J^*(\cdot, Q)$ is lower semicontinuous at P and $|J(P_n, Q, \gamma^*_{P_n}) - J(P, Q, \gamma^*_{P_n})| \to 0$ for every $P_n \in \mathcal{B}^{\rho}_{1/n}(P)$. Then,*

(i) Robustness in the sense of DRO (2.2) implies robustness in the sense of RSP (2.1).

(ii) Suppose further that $Q(dy|x) = f(x, y)\psi(dy)$, and that, under the randomized-policy formulation endowed with the Young topology, the hypotheses of Sion's minimax theorem hold on $\Gamma \times \mathcal{B}^{\rho}_{1/n}(P)$ for every n. Then, robustness in the sense of DRO is equivalent to robustness in the sense of RSP under weak convergence.

Proof. (i) Suppose DRO holds. By Theorem 2.8, we have that $J^*(P_n, Q) \to J^*(P, Q)$. Observe that

$$|J(P, Q, \gamma^*_{P_n}) - J^*(P, Q)| \leq |J(P, Q, \gamma^*_{P_n}) - J(P_n, Q, \gamma^*_{P_n})| + |J^*(P_n, Q) - J^*(P, Q)| \to 0$$

by our assumptions. Thus, RSP holds.

(ii) By Sion's minimax theorem,

$$\inf_{\gamma\in\Gamma} \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma) = \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J^*(\bar{P}, Q).$$

Thus, DRO is equivalent to value convergence under weak convergence. Therefore,

$$|J^*(P_n, Q) - J^*(P, Q)| \leq |J(P_n, Q, \gamma^*_{P_n}) - J(P, Q, \gamma^*_{P_n})| + |J(P, Q, \gamma^*_{P_n}) - J^*(P, Q)|$$

which goes to 0 provided RSP holds. Thus, DRO is equivalent to RSP. ◇

Given the above, we provide sufficient conditions for which DRO, value convergence, and RSP hold simultaneously.

THEOREM 2.10. *Suppose the conditions of Theorem 2.5 hold, then DRO, value convergence, and RSP hold under weak convergence.*

Proof. Let ρ metrize weak convergence. It suffices to show that DRO holds, as taking $P_n \in \mathcal{B}^{\rho}_{1/n}(P)$ trivially satisfies the weak convergence of $P_n \to P$, and thus value convergence by Theorem 2.5, and RSP by Theorem 2.6.

Since $P \in \mathcal{B}^{\rho}_{1/n}(P)$,

$$\inf_{\gamma\in\Gamma} \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma) \geq J^*(P, Q).$$

On the other hand, by choosing γ^*_P,

$$\inf_{\gamma\in\Gamma} \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma) \leq \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma^*_P).$$

Under the conditions of Theorem 2.5, $\int c(x, \gamma^*_P(y))Q(dy|x) \in C_b(\mathbb{X})$. Since ρ metrizes weak convergence,

$$\sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} \left| \int\int c(x, \gamma^*_P(y))Q(dy|x)\bar{P}(dx) - \int\int c(x, \gamma^*_P(y))Q(dy|x)P(dx) \right| \to 0.$$

Therefore, we have

$$\begin{aligned} J^*(P, Q) \leq \inf_{\gamma\in\Gamma} \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma) &\leq \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, Q, \gamma^*_P) - J^*(P, Q) + J^*(P, Q) \\ &\leq \sup_{\bar{P}\in\mathcal{B}^{\rho}_{1/n}(P)} \left| J(\bar{P}, Q, \gamma^*_P) - J^*(P, Q) \right| + J^*(P, Q) \end{aligned}$$

where the last term goes to $J^*(P, Q)$ as $n \to \infty$. Therefore, robustness in the sense of DRO holds. ◇

The above discussion provides insights and conditions that will be useful in the analysis to follow.

3. Dynamic (Multi-Stage) Stochastic Control Optimization. We now consider a multi-stage setup. Let $\mathbb{X}$ denote a Borel space which is the state space of a controlled Markov process and $\mathbb{U}$ is the control action space. Here and throughout the paper $\mathbb{Z}_+$ denotes the set of non-negative integers and $\mathbb{N}$ denotes the set of positive integers.

The update rules of the system are determined by (3.2) and the following: We have $X_0 \sim P$ and

$$\begin{aligned} &\Pr\Big(X_t \in B \,\Big|\, (X, U)_{[0,t-1]} = (x, u)_{[0,t-1]} \Big) \\ &= \int_B \mathcal{T}(dx_t | x_{t-1}, u_{t-1}), B \in \mathcal{B}(\mathbb{X} \times \mathbb{Y}), t \in \mathbb{N}, \end{aligned}$$

where $\mathcal{T}$ is the transition kernel of the model which is a stochastic kernel from $\mathbb{X} \times \mathbb{U}$ to $\mathbb{X}$.

Using stochastic realization results (see [30, Lemma 1.2] or [20, Lemma 3.1]), the process defined above can be represented in functional form as follows:

$$X_{t+1} = f(X_t, U_t, W_t) \tag{3.1}$$

for some measurable functions f, g with $\{W_t\}$ being an independent and identically distributed (i.i.d.) system noise process, which is also independent of X_0. Here, the first equation represents the transition kernel $\mathcal{T}$ as it gives the relation of the most recent state and action variables to the upcoming state. From this representation it can be seen that the probabilistic nature of the kernel is determined by the function f and the probability model of the noise W_t. Let

$$I_t = \{X_{[0,t]}, U_{[0,t-1]}\}, \quad t \in \mathbb{N}, \qquad I_0 = \{X_0\},$$

where

$$U_t = \gamma_t(I_t), \quad t \in \mathbb{Z}_+, \tag{3.2}$$

are the $\mathbb{U}$-valued control actions and

$$X_{[0,t]} = \{X_s,\, 0 \le s \le t\}, \quad U_{[0,t-1]} = \{U_s,\, 0 \le s \le t-1\}.$$

We define Γ_{AD} to be the set of all such admissible deterministic admissible policies. If $\gamma = \{\gamma_t, t \in \mathbb{Z}_+\}$ with $U_t = \gamma_t(X_t)$ for all $t \in \mathbb{Z}_+$, then γ is said to be a deterministic Markov policy and belong to the set of Markov policies Γ_{MD}. If $\gamma = \{\gamma_t, t \in \mathbb{Z}_+\} \in \Gamma_{MD}$ with $\gamma_t \equiv \kappa : \mathbb{X} \to \mathbb{U}$ for some fixed κ, then γ is said to be stationary and we denote by Γ_{SD} the set of all stationary policies. In our paper, the focus will be on Markov and stationary policies (which is without loss for optimal stochastic control, but may be so under the DRO formulation). We will also allow for randomized policies in our analysis, and accordingly in the above the image of γ_t is expanded to be $\mathcal{P}(\mathbb{U})$ valued; in other words (e.g. by [20, Lemma 3.1]) for a Markov policy, $U_t = \gamma_t(x_t, r_t)$ for some independent noise variable r_t; we denote the set of possibly randomized Markov policies by Γ_M and the set of possibly randomized stationary policies by Γ_S.

We let the objective of the agent (decision maker) be the minimization of the infinite horizon discounted cost,

$$J_\beta(P, \mathcal{T}, \gamma) = E_P^{\mathcal{T},\gamma}\left[\sum_{t=0}^{\infty} \beta^t c(X_t, U_t)\right] \tag{3.3}$$

for some discount factor $\beta \in (0,1)$, over the set of admissible policies $\gamma \in \Gamma$, where $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}$ is a Borel-measurable stage-wise cost function and $E_P^{\mathcal{T},\gamma}$ denotes the expectation with initial state probability measure P and transition kernel $\mathcal{T}$ under policy γ. Note that $P \in \mathcal{P}(\mathbb{X})$, where we let $\mathcal{P}(\mathbb{X})$ denote the set of probability measures on $\mathbb{X}$.

We define the optimal cost for the discounted infinite horizon setup as a function of the priors and the transition kernels as

$$J_\beta^*(P, \mathcal{T}) = \inf_{\gamma \in \Gamma} J_\beta(P, \mathcal{T}, \gamma).$$

As in the single-stage problem presented earlier, we compare the two formulations on robustness:

Robustness in Stability to Perturbation (RSP). Let $\{\mathcal{T}_n, n \in \mathbb{N}\}$ be a sequence of transition kernels which converge in some sense to $\mathcal{T}$.

Suppose that $\gamma^*_{\mathcal{T}_n}$ is an optimal policy designed for $\mathcal{T}_n$, and is applied for a true model $\mathcal{T}$. Is it the case that if

$$\mathcal{T}_n, \to \mathcal{T},$$

then

$$J_\beta(P, \mathcal{T}\gamma^*_{\mathcal{T}_n}) \to J_\beta^*(P, \mathcal{T})? \tag{3.4}$$

Value Convergence and Continuity in Models An intermediate question is whether we have convergence of values:

$$J^*_\beta(P, \mathcal{T}_n) \to J^*_\beta(P, \mathcal{T})?$$

Robustness in Terms of (Asymptotic) Distributionally Robust Optimization (DRO)

We note that a mathematical formulation of what it means to be a DRO formulation in a dynamic MDP is non-trivial and there are several distinct formulations (see e.g. [44] where a variety of formulations are presented and compared and [17] who notes that even fundamental problems involving how to formulate distributionally robust Markov decision processes based on the sequentially available information for the leader (minimizer) and the follower (maximizer) are significantly non-trivial). Often, a dynamic programming formulation is imposed apriori: to this end, we should emphasize that in the theory of dynamic games, what the leader knows may lead to formulations where dynamic programming principle may or may not be applicable (see e.g. [68, Section 10.2.6] where distinction between global vs. stage-wise Stackelberg formulations are compared; see also [7, 9]). Under global (Stackelberg equilibrium with Leader Commitment) solution concept, the leader announces the entire collection of policies (over the time horizon) and the follower responds: in this case, the value at a particular time is no longer independent of the history and standard dynamic programming is no longer applicable. Under stage-wise Stackelberg equilibrium, however, one may indeed resort to dynamic programming, and it is this notion that we will pursue in our formulation of DRO in the dynamic setting. In particular, one may also view the problem to be where the leader and the follower utilize stage-wise Markov policies, the leader announces a stage-wise policy and the follower responds given the leader's commitment at the particular time-stage.

Accordingly, a natural candidate is to consider the following, where stationary policies are considered:

$$\lim_{n\to\infty} \left| \inf_{\gamma\in\Gamma_M} \sup_{\bar{\mathcal{T}}:\rho(\bar{\mathcal{T}},\mathcal{T})\le\frac{1}{n}} J_\beta(P, \bar{\mathcal{T}}, c, \gamma) - J^*_\beta(P, \mathcal{T}, c) \right| = 0, \tag{3.5}$$

or a generalization where the leader and the follower are allowed to include time-dependent (Markov) policies. A corollary of our analysis via dynamic programming will be to show that a time-dependent relaxation is redundant in this case, and the above does capture a DRO formulation even when the maximizer is allowed to select the kernel depending on the time stage.

3.1. Dynamic Counterexamples to Equivalence. We once again show two examples to highlight that there is no implication relationship between the two robustness formulations in the multi-stage dynamic setting in general. Both examples follow directly from the single-stage setup

EXAMPLE 3.1.

In the following, we will show that robustness in the sense of RSP does not imply robustness in the sense of DRO.

Let the state space be $\mathbb{X} = \mathbb{R}^2$*, where the state at time* t *is given by* $X_t := (R_t, Z_t)$*. Let* $\mathbb{U} = \mathbb{R}$ *denote the action space.*

Let for $x_t \in \mathbb{R}^2$*,* $Q(dy_t|x_t) = \mathrm{Unif}([-1, 1])$*, so that* Q *is a non-informative channel.*

Let the stage-wise cost function be given as:

$$c(X_t, U_t) = R_t^2 e^{R_t^2}(Z_t - U_t)^2$$

We let $\mathcal{T}_n$ *be defined such that* X_{t+1} *evolves independently of the action* U_t *as follows:*

$$R_{t+1} \sim (1 - \frac{1}{n})\delta_0 + \frac{1}{n}\mathcal{N}(0, 1)$$

$$Z_{t+1} \sim \delta_{\frac{1}{n}}$$

where $\mathcal{N}(0,1)$ is a standard Gaussian with mean 0 and unit variance. We let $P = \delta_{(0,0)}$. As $n \to \infty$, it is clear that $\mathcal{T}_n \to \mathcal{T}$ weakly, where the transition kernel $\mathcal{T}$ is $\delta_{(0,0)}$. Thus, we have the following cost criterion:

$$J_\beta(P, \mathcal{T}_n, \gamma) = E_P^{\mathcal{T}_n, \gamma}[\sum_{t=0}^{\infty} \beta^t R_t^2 e^{R_t^2}(Z_t - U_t)^2]$$

Under $\mathcal{T}_n$, the optimal policy $\gamma^*_{\mathcal{T}_n}(Y_t) = \frac{1}{n}$ for all $t \in \mathbb{Z}_+$, since $Z_t = \frac{1}{n}$, almost surely. Thus, we have the following.

$$J_\beta(P, \mathcal{T}_n, \gamma^*_{\mathcal{T}_n}) = E_P^{\mathcal{T}_n, \gamma^*_{\mathcal{T}_n}}[\sum_{t=0}^{\infty} \beta^t R_t^2 e^{R_t^2}(0)^2] = 0$$

If we then apply this constructed policy $\gamma^*_{\mathcal{T}_n}$ to the true model $\mathcal{T}$, the cost is:

$$J_\beta(P, \mathcal{T}, \gamma^*_{\mathcal{T}_n}) = E_P^{\mathcal{T}, \gamma^*_{\mathcal{T}_n}}[\sum_{t=0}^{\infty} \beta^t (0)^2 e^{(0)^2}(0 - \frac{1}{n})^2] = 0$$

Since the optimal cost for the true model is $J^*_\beta(P, \mathcal{T}) = 0$, we have that

$$\lim_{n\to\infty} \left| J_\beta(P, \mathcal{T}, \gamma^*_{\mathcal{T}_n}) - J^*_\beta(P, \mathcal{T}) \right| = 0$$

Under DRO, however; for $n \in \mathbb{N}$ we have that the ambiguity set contains adversarial kernels $\bar{\mathcal{T}}$ where Z_t is no longer a deterministic constant. Because Q is non-informative, any policy $\gamma \in \Gamma$ either outputs a deterministic value for U_t, or a randomized output that is uninformative about the state. Thus, we have that $(Z_t - U_t)^2 > 0$ on a set of positive measure. Since $R_t^2 e^{R_t^2}$ is not integrable, the expectation diverges for any $\gamma \in \Gamma$. Therefore, for all $n \in \mathbb{N}$, we have

$$\inf_{\gamma\in\Gamma} \sup_{\rho(\bar{\mathcal{T}}, \mathcal{T}) \le \frac{1}{n}} J_\beta(P, \bar{\mathcal{T}}, \gamma) = \infty$$

EXAMPLE 3.2. *In the following, we will show that robustness in the sense of DRO does not imply robustness in the sense of RSP.*

Let $\beta \in (0,1)$ be fixed. Let $\mathbb{U} = [0,1]$ denote the action space and $\mathbb{X} = \{-1, 0\} \cup \{1/n : n \in \mathbb{N}\}$ denote the state space. Let for $x_t \in \mathbb{X}$, $Q(dy_t|x_t) = \mathrm{Unif}([-1,1])$, so that Q is a non-informative channel.

Let $P = \delta_{-1}$ be fixed. Define the nominal transition kernel $\mathcal{T}$ by $\mathcal{T}(\cdot|x,u) = \delta_0$ for every $(x,u) \in \mathbb{X}\times\mathbb{U}$, and for $n \in \mathbb{N}$ define the perturbed transition kernel $\mathcal{T}_n$ by $\mathcal{T}_n(\cdot|x,u) = \delta_{1/n}$. Endow $\mathcal{P}(\mathbb{X})$ with any metric ρ which metrizes weak convergence and define

$$\rho_\infty(\mathcal{T}_1, \mathcal{T}_2) = \sup_{(x,u)\in\mathbb{X}\times\mathbb{U}} \rho\big(\mathcal{T}_1(\cdot|x,u), \mathcal{T}_2(\cdot|x,u)\big).$$

Since $\delta_{1/n} \to \delta_0$ weakly, we have $\rho_\infty(\mathcal{T}_n, \mathcal{T}) = \rho(\delta_{1/n}, \delta_0) \to 0$. Let the stage-wise cost function $c : \mathbb{X}\times\mathbb{U} \to \mathbb{R}_+$ be given by

$$c(-1,u) = 0, \qquad c(0,u) = u^2$$

and, for $n \in \mathbb{N}$,

$$c(1/n, u) = \frac{1}{n}(1-u)^2.$$

Under the perturbed model $(P, \mathcal{T}_n)$*, we have* $X_0 = -1$ *and* $X_t = 1/n$ *for every* $t \geq 1$*. Thus, the stationary policy* $\gamma^*_{\mathcal{T}_n}(Y_{[0,t]}, U_{[0,t-1]}) = 1$ *for every* $t \geq 0$ *is optimal, and* $J^*_\beta(P, \mathcal{T}_n) = J_\beta(P, \mathcal{T}_n, \gamma^*_{\mathcal{T}_n}) = 0$*. For the true model* $(P, \mathcal{T})$*, the policy taking* $U_t = 0$ *for every* $t \geq 0$ *has zero cost, and hence* $J^*_\beta(P, \mathcal{T}) = 0$*. However, when applying the optimal policy of the perturbed model to the true model, we have* $X_0 = -1$ *and* $X_t = 0$ *for every* $t \geq 1$*. Therefore,*

$$J_\beta(P, \mathcal{T}, \gamma^*_{\mathcal{T}_n}) = E_P^{\mathcal{T}, \gamma^*_{\mathcal{T}_n}}\left[\sum_{t=0}^{\infty} \beta^t c(X_t, U_t)\right] = \sum_{t=1}^{\infty} \beta^t = \frac{\beta}{1-\beta} \neq 0.$$

Thus, robustness in the sense of RSP does not hold. That is,

$$J_\beta(P, \mathcal{T}, \gamma^*_{\mathcal{T}_n}) \not\to J^*_\beta(P, \mathcal{T}).$$

We now show that the DRO criterion holds. We must show that

$$\inf_{\gamma \in \Gamma} \sup_{\bar{\mathcal{T}}: \rho_\infty(\bar{\mathcal{T}}, \mathcal{T}) \leq r} J_\beta(P, \bar{\mathcal{T}}, \gamma) \to 0 = J^*_\beta(P, \mathcal{T}).$$

We obtain an upper bound by choosing the admissible policy $\gamma_t(Y_{[0,t]}, U_{[0,t-1]}) = 0$ *for* $t \geq 0$*. Let* $g(x) = c(x, 0)$*, so that*

$$g(-1) = g(0) = 0, \qquad g(1/n) = \frac{1}{n},$$

and hence $g \in C_b(\mathbb{X})$*.*

Since ρ *metrizes weak convergence and* $g \in C_b(\mathbb{X})$*, we have*

$$\sup_{\mu: \rho(\mu, \delta_0) \leq r} \int g(x)\mu(dx) \to 0$$

as $r \to 0$*.*

For every $\bar{\mathcal{T}}$ *such that* $\rho_\infty(\bar{\mathcal{T}}, \mathcal{T}) \leq r$*, we have* $\rho\big(\bar{\mathcal{T}}(\cdot|x, u), \delta_0\big) \leq r$ *for every* $(x, u) \in \mathbb{X} \times \mathbb{U}$*. Thus, under the policy* $U_t = 0$*,*

$$E_P^{\bar{\mathcal{T}}, 0}[g(X_t)] \leq \sup_{\mu: \rho(\mu, \delta_0) \leq r} \int g(x)\mu(dx)$$

for every $t \geq 1$*. Therefore,*

$$\begin{aligned} 0 \leq \inf_{\gamma \in \Gamma} \sup_{\bar{\mathcal{T}}: \rho_\infty(\bar{\mathcal{T}}, \mathcal{T}) \leq r} J_\beta(P, \bar{\mathcal{T}}, \gamma) &\leq \sup_{\bar{\mathcal{T}}: \rho_\infty(\bar{\mathcal{T}}, \mathcal{T}) \leq r} J_\beta(P, \bar{\mathcal{T}}, 0) \\ &\leq \sum_{t=1}^{\infty} \beta^t \sup_{\mu: \rho(\mu, \delta_0) \leq r} \int g(x)\mu(dx) \\ &= \frac{\beta}{1-\beta} \sup_{\mu: \rho(\mu, \delta_0) \leq r} \int g(x)\mu(dx) \to 0. \end{aligned}$$

Therefore, we have that

$$\inf_{\gamma\in\Gamma}\sup_{\bar{\mathcal{T}}:\rho_\infty(\bar{\mathcal{T}},\mathcal{T})\le r} J_\beta(P,\bar{\mathcal{T}},\gamma)\to J^*_\beta(P,\mathcal{T}),$$

that is,

$$\lim_{n\to\infty}\left|\inf_{\gamma\in\Gamma}\sup_{\bar{\mathcal{T}}:\rho_\infty(\bar{\mathcal{T}},\mathcal{T})\le 1/n} J_\beta(P,\bar{\mathcal{T}},\gamma)-J^*_\beta(P,\mathcal{T})\right|=0.$$

Thus, robustness in the sense of DRO holds while robustness in the sense of RSP does not.

THEOREM 3.1. *In the dynamic setup, we have that in general,*

(i) Robustness in the sense of RSP (3.4) does not imply robustness in the sense of DRO (3.5), and

(ii) Robustness in the sense of DRO (3.5) does not imply robustness in the sense of RSP (3.4).

Proof. Part (i) follows from Example 3.1, while part (ii) follows from Example 3.2. ◇

3.2. Equivalence of RSP and DRO in the Dynamic Formulation. The following section provides abstract conditions for which the formulations of robustness are equivalent, after which we provide an explicit construction for which all robustness formulations hold in the weak convergence and TV cases respectively.

3.2.1. Dynamic DRO Convergence and Value Convergence. Under the preceding formulation of DRO, we can extract similar implications on the optimal value function as we did in the single-stage static case.

The following results are analogous to the single-stage case. In the first lemma, we suppose that the ambiguity sets shrink with respect to some metric ρ as $n\to\infty$. We will see later that the notion of convergence given from the formulation of the ambiguity sets will automatically give us value convergence and RSP robustness. Thus, we do not restrict our initial choice of metric ρ.

LEMMA 3.2. *Suppose that robustness in the sense of DRO (3.5) holds for a sequence of shrinking ambiguity sets $\mathcal{B}^\rho_{1/n}(\mathcal{T})$ defined with respect to some metric ρ and containing $\mathcal{T}$. Then, DRO robustness implies upper semicontinuity of the optimal discounted value at $\mathcal{T}$ along the ambiguity sets. That is, $J^*_\beta(P,\cdot,c)$ is upper semicontinuous.*

Proof. By the minimax inequality,

$$\inf_{\gamma\in\Gamma}\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J_\beta(P,\mathcal{R},c,\gamma)\ge\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})}\inf_{\gamma\in\Gamma} J_\beta(P,\mathcal{R},c,\gamma)=\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J^*_\beta(P,\mathcal{R},c).$$

Since $\mathcal{T}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})$,

$$J^*_\beta(P,\mathcal{T},c)\le\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J^*_\beta(P,\mathcal{R},c).$$

Therefore, DRO robustness implies

$$\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J^*_\beta(P,\mathcal{R},c)\to J^*_\beta(P,\mathcal{T},c).$$

Suppose $\mathcal{R}_n\to\mathcal{T}$ under ρ. For every $k\in\mathbb{N}$, there exists N such that, for all $n\ge N$, $\mathcal{R}_n\in\mathcal{B}^\rho_{1/k}(\mathcal{T})$. Therefore,

$$\limsup_{n\to\infty} J^*_\beta(P,\mathcal{R}_n,c)\le\sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/k}(\mathcal{T})} J^*_\beta(P,\mathcal{R},c).$$

Taking $k \to \infty$ gives

$$\limsup_{n\to\infty} J^*_\beta(P, \mathcal{R}_n, c) \le J^*_\beta(P, \mathcal{T}, c).$$

Thus, $J^*_\beta(P, \cdot, c)$ is upper semicontinuous at $\mathcal{T}$. ◇

THEOREM 3.3. *Suppose that robustness in the sense of DRO (3.5) holds and that $J^*_\beta(P, \cdot, c)$ is lower semicontinuous at $\mathcal{T}$. Then,*

$$J^*_\beta(P, \mathcal{R}_n, c) \to J^*_\beta(P, \mathcal{T}, c)$$

for every $\mathcal{R}_n \in \mathcal{B}^\rho_{1/n}(\mathcal{T})$.

Proof. Suppose DRO holds. Let $\mathcal{R}_n \in \mathcal{B}^\rho_{1/n}(\mathcal{T})$. As $\mathcal{R}_n \to \mathcal{T}$ under ρ, and from Lemma 3.2 and the lower semicontinuity of $J^*_\beta(P, \cdot, c)$, we have that

$$J^*_\beta(P, \mathcal{T}, c) \le \liminf_{n\to\infty} J^*_\beta(P, \mathcal{R}_n, c) \le \limsup_{n\to\infty} J^*_\beta(P, \mathcal{R}_n, c) \le J^*_\beta(P, \mathcal{T}, c).$$

Thus, this gives us a similar notion of value convergence for the perturbed models $\mathcal{R}_n$. ◇

3.2.2. Equivalence of RSP and DRO. Before we provide sufficient conditions for which the robustness formulations hold simultaneously, we provide a similar argument to Theorem 2.9, but for the dynamic case. Note, using Sion's minimax theorem in the global problem is non-trivial and requires much greater assumptions than in the single-stage case. In the following theorem, we initially assume a minimax identity which we will later establish under stage-wise minimax interchange together with rectangularity, measurable selection and gluing, and dynamic programming.

THEOREM 3.4. *Suppose that $J^*_\beta(P, \cdot, c)$ is lower semicontinuous at $\mathcal{T}$ and $|J_\beta(P, \mathcal{R}_n, c, \gamma^*_{\mathcal{R}_n}) - J_\beta(P, \mathcal{T}, c, \gamma^*_{\mathcal{R}_n})| \to 0$ for every $\mathcal{R}_n \in \mathcal{B}^\rho_{1/n}(\mathcal{T})$. Then,*

(i) Robustness in the sense of DRO (3.5) implies robustness in the sense of RSP (3.4).

(ii) Suppose further that

$$\inf_{\gamma\in\Gamma} \sup_{\mathcal{R}\in B^\rho_{1/n}(\mathcal{T})} J_\beta(P, \mathcal{R}, c, \gamma) = \sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} \inf_{\gamma\in\Gamma} J_\beta(P, \mathcal{R}, c, \gamma)$$

for every n. Then, robustness in the sense of DRO is equivalent to robustness in the sense of RSP under ρ.

Proof. (i) Suppose DRO holds. By Theorem 3.3, $J^*_\beta(P, \mathcal{R}_n, c) \to J^*_\beta(P, \mathcal{T}, c)$. Thus,

$$\begin{aligned}|J_\beta(P, \mathcal{T}, c, \gamma^*_{\mathcal{R}_n}) - J^*_\beta(P, \mathcal{T}, c)| \le& |J_\beta(P, \mathcal{T}, c, \gamma^*_{\mathcal{R}_n}) - J_\beta(P, \mathcal{R}_n, c, \gamma^*_{\mathcal{R}_n})| \\ &+ |J^*_\beta(P, \mathcal{R}_n, c) - J^*_\beta(P, \mathcal{T}, c)| \to 0\end{aligned}$$

by our assumption. Thus, RSP holds.

(ii) By the minimax identity, we have,

$$\inf_{\gamma\in\Gamma} \sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J_\beta(P, \mathcal{R}, c, \gamma) = \sup_{\mathcal{R}\in\mathcal{B}^\rho_{1/n}(\mathcal{T})} J^*_\beta(P, \mathcal{R}, c).$$

Therefore,

$$\begin{aligned}|J^*_\beta(P, \mathcal{R}_n, c) - J^*_\beta(P, \mathcal{T}, c)| \le& |J_\beta(P, \mathcal{R}_n, c, \gamma^*_{\mathcal{R}_n}) - J_\beta(P, \mathcal{T}, c, \gamma^*_{\mathcal{R}_n})| \\ &+ |J_\beta(P, \mathcal{T}, c, \gamma^*_{\mathcal{R}_n}) - J^*_\beta(P, \mathcal{T}, c)| \to 0\end{aligned}$$

which goes to 0 provided RSP holds. Thus, DRO is equivalent to RSP. ◇

Given the above analysis, the following sections are aimed at providing explicit conditions for which DRO, value convergence, and RSP robustness hold. That is, the following sections provide sufficient conditions for which the abstract hypotheses are satisfied.

3.2.3. Robustness in the Sense of RSP under Continuous Weak Convergence. We start with the following assumption.

ASSUMPTION 3.1.

(a) [Continuous Weak Convergence of Kernels] The sequence of transition kernels $\mathcal{T}_n$ satisfies the following: $\{\mathcal{T}_n(\cdot|x_n,u_n), n\in\mathbb{N}\}$ converges weakly to $\mathcal{T}(\cdot|x,u)$ for any sequence $\{x_n,u_n\}\subset\mathbb{X}\times\mathbb{U}$ and $x,u\in\mathbb{X}\times\mathbb{U}$ such that $(x_n,u_n)\to(x,u)$.

(b) The stochastic kernels $\mathcal{T}(\cdot|x,u)$ and $\{\mathcal{T}_n(\cdot|x,u)\}_n$ are weakly continuous in (x,u).

(c) The stage-wise cost function $c(x,u)$ is non-negative, bounded, and continuous on $\mathbb{X}\times\mathbb{U}$.

(d) $\mathbb{U}$ is compact.

We emphasize the importance of continuous weak convergence, and inadequacy of pointwise weak convergence.

THEOREM 3.5. *[41] For $\mathcal{T}_n\to\mathcal{T}$ weakly, it is not necessarily true that $J_\beta(\mathcal{T}_n,\gamma^*_{\mathcal{T}_n})\to J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}})$ even when the prior distributions are the same and $c(x,u)$ is continuous and bounded in $\mathbb{X}\times\mathbb{U}$.*

THEOREM 3.6. *[41] Supposing $\mathcal{T}_n(\cdot|x_n,u_n)\to\mathcal{T}(\cdot|x,u)$ weakly for every $x\in\mathbb{X}$ and $u\in\mathbb{U}$ and $(x_n,u_n)\to(x,u)$, then it is not true in general that $J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}_n})\to J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}})$ (that is, (3.4) may not hold), even when $\mathbb{X}$ and $\mathbb{U}$ are compact and $c(x,u)$ is continuous and bounded in $\mathbb{X}\times\mathbb{U}$.*

However, we have the positive result under continuous weak convergence:

THEOREM 3.7. *[41] Under Assumption 3.1, $J_\beta(\mathcal{T}_n,\gamma^*_{\mathcal{T}_n})\to J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}})$ for any initial state x_0, as $n\to\infty$.*

We now present a RSP robustness result: If the optimal policies for every initial point are identical in the sense that, for every n, γ^*_n is optimal for every $x_0\in\mathbb{X}$ (under the model $\mathcal{T}_n$). A sufficient condition for this property is that γ^*_n solves the discounted cost optimality equation (DCOE) for every initial point.

A policy $\gamma^*\in\Gamma$ solves the discounted cost optimality equation and is optimal if it satisfies

$$J^*_\beta(\mathcal{T},x)=c(x,\gamma^*(x))+\beta\int J^*_\beta(\mathcal{T},x_1)\mathcal{T}(dx_1|x,\gamma^*(x)). \tag{3.6}$$

Thus, a policy is optimal for every initial point if it satisfies the DCOE for all initial points $x\in\mathbb{X}$.

THEOREM 3.8. *[41] [RSP Robustness under Continuous weak Convergence] Under Assumption 3.1, (3.4) holds, that is $J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}_n})\to J_\beta(\mathcal{T},\gamma^*_{\mathcal{T}})$ for any initial point x_0 if $\gamma^*_{\mathcal{T}_n}$ is optimal for any initial point for the kernel $\mathcal{T}_n$ (in the sense of satisfying (3.6)).*

3.2.4. Robustness under DRO. To be able to apply Sion's minimax theorem, one should allow for randomized policies unless convexity in actions is imposed during dynamic programming; we refer the reader to [13] which considered a more restrictive model on the ambiguity set and accordingly imposed convexity of the value functions in the deterministic actions; which, however, limits the applicability; see [13, Lemma 4 and Remark 3]. Note also that the setup below is different with respect to the formulation in [13] which considered ambiguity in the distribution of a driving noise and the functional form (or the stochastic realization form of a kernel; see [4, Lemma F], [30, Lemma 1.2], or [20, Lemma 3.1]) was taken to be known and fixed.

In the following, we assume that $\mathcal{T}$ and the approximating kernels $\mathcal{R}$ are weakly continuous kernels.

In our case, we allow for randomized policies and also require that the set of allowable transition kernels at any state x, denoted with $\mathcal{D}_x$, be compact which in turn imposes an equi-continuity conditions on $\{\mathcal{T}(\cdot|x,\cdot):\mathbb{U}\to\mathcal{P}(\mathbb{X})\}\in\mathcal{D}_x$ so that this family is weakly compact by an application of a generalized Arzela-Ascoli theorem [50, Theorem 1.6.14] (see [56, Theorem 4.2 and Remark 8]).

$$\min_{\gamma(du|x)\in\mathcal{P}(\mathbb{U})}\sup_{\mathcal{R}\in\mathcal{D}_x^n}\int\left(c(x,u)+\beta\int\mathcal{R}(dx_{t+1}|x_t=x,u_t=u)J(x_{t+1})\right)\gamma(du|x),\quad(3.7)$$

where

$$\mathcal{D}_x^n:=\left\{\mathcal{R}:\sup_{u\in\mathbb{U}}\rho(\mathcal{R}(\cdot|x,u),\mathcal{T}(\cdot|x,u))\leq\frac{1}{n}\right\}\quad(3.8)$$

We recall the following [45, Theorem 2.12].

LEMMA 3.9. *If $\mathbb{U}\ni u\mapsto\mathcal{R}(\cdot|x,\cdot)\in\mathcal{P}(\mathbb{X})$ is uniformly equicontinuous, $\mathbb{U}$ compact and $\mathcal{P}(\mathbb{X})$ is compact (under weak convergence topology), then $\mathcal{D}_x^n$ is weakly pre-compact.*

Since $\mathcal{P}(\mathbb{X})$ is compact every sequence $\mathcal{R}(\cdot|x,u_n)$ would have a converging subsequence which would be a probability measure, it follows that $\mathcal{D}_x^n$ is closed and we can then strengthen the result above to $\mathcal{D}_x^n$ being compact.

We have that (i) for each fixed x, the spaces of policies and perturbations are convex and compact, (ii) the integrand is linear both in $\gamma(\cdot|x)$ and $\mathcal{R}(\cdot|x,u)$ when either other one is fixed, and (iii) also continuous in either one if J is, and if $\mathcal{R}(\cdot|x,u)$ is a weakly continuous kernel. Accordingly, The order of $\min$ and $\max$ can be swapped, and then the analysis of

$$\sup_{\mathcal{R}\in\mathcal{D}_x^n}\min_{\gamma(du|x)\in\mathcal{P}(\mathbb{U})}\int\left(c(x,u)+\beta\int\mathcal{R}(dx_{t+1}|x_t=x,u_t=u)J(x_{t+1})\right)\gamma(du|x),$$

becomes an instance of that for value convergence when kernels converge pointwise weakly.

However, as we observed in Theorem 3.6, pointwise convergence is not sufficient and continuous weak convergence is needed in view of Theorem 3.8. Accordingly, we will consider a more restrictive ambiguity class

$$\mathcal{D}'^n:=\left\{\mathcal{R}:\sup_{(x,u)\in\mathbb{X}\times\mathbb{U}}\rho(\mathcal{R}(\cdot|x,u),\mathcal{T}(\cdot|x,u))\leq\frac{1}{n}\right\}\quad(3.9)$$

which in turn implies continuous weak convergence [56, Section 5.1.1] or [45, Proposition 2.8] as $n\to\infty$, given that $\mathcal{T}$ is weakly continuous.

In this case, the DRO formulation is to be written as

$$\min_{\gamma(du|x)\in\mathcal{P}(\mathbb{U})}\sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}_x^n\cap\mathcal{D}'^n}\int\left(c(x,u)+\beta\int\mathcal{R}(dx_{t+1}|x_t=x,u_t=u)J(x_{t+1})\right)\gamma(du|x),\quad(3.10)$$

Finally, to obtain a notationally less demanding formulation, we instead consider the following ambiguity class:

$$\mathcal{D}''^n_x:=\Big\{\mathcal{R}(\cdot|x,\cdot)\in C(\mathbb{U},\mathcal{P}(\mathbb{X})):\ \sup_{u\in\mathbb{U}}\rho(\mathcal{R}(\cdot|x,u),\mathcal{T}(\cdot|x,u))\leq\frac{1}{n},\ u\mapsto\mathcal{R}(\cdot|x,u)\text{ is uniformly equicontinuous}\Big\},\quad(3.11)$$

where the ambiguity class of global kernels is defined as

$$\mathcal{D}''^n:=\Big\{\mathcal{R}:\ x\mapsto\mathcal{R}(\cdot|x,\cdot)\text{ is measurable},\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x\text{ for every }x\in\mathbb{X}\Big\}.\quad(3.12)$$

It follows that ${\mathcal{D}''}^n_x$ is a compact set of kernels under the supremum norm (where the pointwise convergence is under weak convergence). For ${\mathcal{D}''}^n$, convergence in this norm implies continuous weak convergence [56, Section 5.1.1] or [45, Proposition 2.8] when $\mathcal{T}$ is weak Feller.

Choice of metrics and convexity of the ambiguity sets. An arbitrary metric may not have convex metric balls; accordingly, throughout this section we restrict our attention to integral probability metrics on $\mathcal{P}(\mathbb{X})$ of the form

$$\rho_{\mathcal{F}}(\pi_1, \pi_2) := \sup_{f \in \mathcal{F}} \left| \int f d\pi_1 - \int f d\pi_2 \right|,$$

where $\mathcal{F}$ is some function class. We have that these balls are indeed convex as, for $\mu \in \mathcal{P}(\mathbb{X})$, $r > 0$, $\pi_1, \pi_2 \in \mathcal{B}_r^{\rho_{\mathcal{F}}}(\mu)$, and $\lambda \in [0, 1]$, we have

$$\begin{aligned}
\rho_{\mathcal{F}}(\lambda\pi_1 + (1-\lambda)\pi_2, \mu) &= \sup_{f \in \mathcal{F}} \left| \lambda \left(\int f d\pi_1 - \int f \mu \right) + (1-\lambda) \left(\int f d\pi_2 - \int f d\mu \right) \right| \\
&\leq \lambda \rho_{\mathcal{F}}(\pi_1, \mu) + (1-\lambda)\rho_{\mathcal{F}}(\pi_2, \mu) \\
&\leq r
\end{aligned} \tag{3.13}$$

So, if the weak topology metric is defined as such (e.g. bounded-Lipschitz metric where $\mathcal{F} = \{f : ||f||_{BL} \leq 1\}$), they will have convex ambiguity sets. The same convexity conclusion holds for total variation.

Consider then the following value iteration equations. Let $J_0^n \equiv 0$ and for $m > 0$:

$$\begin{aligned}
&J_{m+1}^n(x) \\
&= \min_{\gamma(du|x) \in \mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot) \in {\mathcal{D}''}^n_x} \int \left(c(x,u) + \beta \int \mathcal{R}(dx_{t+1}|x_t = x, u_t = u) J_n^m(x_{t+1}) \right) \gamma(du|x)
\end{aligned} \tag{3.14}$$

For the rest of this section, we will impose the following assumption.

ASSUMPTION 3.2. *Suppose that $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}$ is continuous and bounded, and let $\mathcal{T}$ be a weakly continuous stochastic kernel. Let $\rho_{\mathcal{F}}$, from above, be a metric on $\mathcal{P}(\mathbb{X})$ that metrizes weak convergence. Let the uncertainty sets be characterized by $\mathcal{R} \in {\mathcal{D}''}^n$, where each $\mathcal{R}$ is also weakly continuous, and $x \mapsto {\mathcal{D}''}^n_x$ is upper semi-continuous.*

To simplify some of the notation, for $(x, \gamma, q) \in \mathbb{X} \times \mathcal{P}(\mathbb{U}) \times {\mathcal{D}''}^n_x$, where $\mathcal{P}(\mathbb{U})$ is endowed with the weak topology and ${\mathcal{D}''}^n_x$ with the supremum topology, define

$$g_J(x, \gamma, q) := \int \left(c(x,u) + \beta \int J(x') q(dx'|u) \right) \gamma(du|x). \tag{3.15}$$

As $q \in {\mathcal{D}''}^n_x$ is dependent on x, we will sometimes use $\mathcal{R}(\cdot|x,\cdot) := q(\cdot|\cdot)$.

PROPOSITION 3.10. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.2 holds. Then, the iterations defined by (3.14) are well-defined and define a contraction with contraction rate β. Therefore, the iterations converge to a fixed point $J_\infty^n \in C_b(\mathbb{X})$ and satisfy*

$$\lim_{m \to \infty} ||J_m^n - J_\infty^n||_\infty = 0.$$

Proof. We first note that the compactness and convexity of ${\mathcal{D}''}^n_x$ follows directly from Lemma 3.9 and from our choice of convex metrics.

Now, observe that for a continuous and bounded function $g : \mathbb{X}_1 \times \mathbb{X}_2 \times \mathbb{X}_3(x) \ni (x, u, v) \mapsto g(x, u, v) \in \mathbb{R}$, if $\mathbb{X}_2$ is compact and $\mathbb{X}_3(x) : x \to 2^{\mathbb{X}_3}$ is upper semi-continuous

and compact valued, then $G(x) := \min_{u\in\mathbb{X}_2}\max_{v\in\mathbb{X}_3(x)} g(x,u,v)$ is continuous in x (see e.g. [67, Lemma 5.2] for this standard result). We have that $\mu \mapsto \int J d\mu$ is continuous under weak convergence and $x \mapsto \mathcal{D}''^n_x$ is continuous. Thus, g_J (3.15) is jointly continuous in (x,γ,q). By the compactness of $\mathcal{P}(\mathbb{U})$ and $\mathcal{D}''^n_x$, we have for $J^n_m \in C_b(\mathbb{X})$,

$$J^n_{m+1}(x) = \min_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{q\in\mathcal{D}''^n_x} g_{J^n_m}(x,\gamma,q) \in C_b(\mathbb{X}).$$

Now, observe that the iterations defined by (3.14) are well-defined and define a contraction with contraction rate β which follows from the argument that for any $J^n_m, \bar{J}^n_m \in C_b(\mathbb{X})$:

$$\begin{aligned}
&\Big| \min_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) J^n_m(x') \Big)\gamma(du|x) \\
&\quad - \min_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) \bar{J}^n_m(x') \Big)\gamma(du|x) \Big| \\
&\leq \max_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \Big| \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) J^n_m(x') \Big)\gamma(du|x) \\
&\qquad\qquad - \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) \bar{J}^n_m(x') \Big)\gamma(du|x) \Big| \\
&\leq \max_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \Big| \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) J^n_m(x') \Big)\gamma(du|x) \\
&\qquad\qquad - \int \Big(c(x,u) + \beta \int \mathcal{R}(dx'|x,u) \bar{J}^n_m(x') \Big)\gamma(du|x) \Big| \\
&= \max_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \Big| \int \beta \int \mathcal{R}(dx'|x,u) J^n_m(x') \Big)\gamma(du|x) \\
&\qquad\qquad - \int \beta \int \mathcal{R}(dx'|x,u) \bar{J}^n_m(x') \Big)\gamma(du|x) \Big| \\
&\leq \beta ||J^n_m - \bar{J}^n_m||_\infty. \qquad (3.16)
\end{aligned}$$

That is, $||J^n_{m+1} - \bar{J}^n_{m+1}||_\infty \leq \beta ||J^n_m - \bar{J}^n_m||_\infty$. Therefore, the iterations converge to a fixed point $J^n_\infty \in C_b(\mathbb{X})$ and satisfy

$$\lim_{m\to\infty} ||J^n_m - J^n_\infty||_\infty = 0.$$

◇

PROPOSITION 3.11. *Suppose Assumption (3.2) holds. Then, for $J \in C_b(\mathbb{X})$,*

$$\min_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{q\in\mathcal{D}''^n_x} g_J(x,\gamma,q) = \sup_{q\in\mathcal{D}''^n_x} \min_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} g_J(x,\gamma,q).$$

Proof. As g_J (3.15) is affine in γ and q, and since $\mathcal{P}(\mathbb{U})$ and $\mathcal{D}''^n_x$ are compact and convex, Sion's minimax theorem gives the result. ◇

THEOREM 3.12. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.2 holds. Then, $J^n_\infty(x)$ is indeed the DRO value for $\mathcal{D}''^n$. That is,*

$$J^n_\infty(x) = \inf_{\gamma\in\Gamma} \sup_{\mathcal{R}\in\mathcal{D}''^n} J_\beta(x,\mathcal{R},c,\gamma).$$

Proof. Now, for every x, Sion's theorem and compactness gives a saddle-point $(\gamma^*_n(\cdot|x), q^*_n(x)) \in \mathcal{P}(\mathbb{U}) \times \mathcal{D}''^n_x$ for $g_{J^n_\infty}(x,\cdot,\cdot)$. That is,

$$g_{J^n_\infty}(x,\gamma^*_n(\cdot|x),q) \leq J^n_\infty(x) \leq g_{J^n_\infty}(x,\gamma,q^*_n(x))$$

for every $\gamma \in \mathcal{P}(\mathbb{U})$ and every $q \in \mathcal{D}''^n_x$, and where $g_{J^n_\infty}(x, \gamma^*_n(\cdot|x), q^*_n(x)) = J^n_\infty(x)$. A measurable selection theorem (see [54, Theorem 4.2, Theorem 4.3]) yields measurable selectors for $x \mapsto \gamma^*_n(\cdot|x)$ and $x \mapsto q^*_n(x)$. We define the glued kernel $\mathcal{R}^*_n(\cdot|x,u) := q^*_n(x)(\cdot|u)$. Indeed, $\mathcal{R}^*_n$ is a measurable stochastic kernel, and thus $\mathcal{R}^*_n \in \mathcal{D}''^n$.

From the saddle-point inequalities, we can establish that $J^n_\infty(x)$ is indeed the DRO value for $\mathcal{D}''^n$. That is,

$$J^n_\infty(x) = \inf_{\gamma\in\Gamma} \sup_{\mathcal{R}\in\mathcal{D}''^n} J_\beta(x, \mathcal{R}, c, \gamma).$$

◇

THEOREM 3.13. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.2 holds. The DRO solution can be obtained via the iterations given by (3.14) so that*

$$\lim_{n\to\infty} \lim_{m\to\infty} J^n_m(x) = J^*_\beta(x).$$

Proof. Now, we would like to show convergence of the DRO values. Formally, we wish to show that for

$$J^*_\beta(x) := J^\infty_\infty(x)$$

that

$$|J^n_\infty(x) - J^\infty_\infty(x)| \to 0.$$

We have that

$$|J^n_\infty(x) - J^\infty_\infty(x)| \le |J^n_\infty(x) - J^n_m(x)| + |J^n_m(x) - J^\infty_m(x)| + |J^\infty_m(x) - J^\infty_\infty(x)|$$

where $J^\infty_m(x)$ is the value iteration applied for the nominal model.

By iterated application of convergence with $J^n_0 \equiv 0$ and $J^\infty_0 \equiv 0$, for every fixed m, it can be shown that

$$|J^n_m(x) - J^\infty_m(x)| \to 0,$$

as $n \to \infty$. Furthermore, the first and the third terms above can be made arbitrarily small by taking m large enough. Therefore, for every $\epsilon > 0$, there exists $\bar{m}$ so that $|J^n_\infty(x) - J^n_m(x)| \le \epsilon/3$ and $|J^\infty_m(x) - J^\infty(x)| \le \epsilon/3$ for all $m \ge \bar{m}$ and for this $\bar{m}$ we have that there exists n large enough that $|J^n_{\bar{m}}(x) - J^\infty_{\bar{m}}(x)| \le \epsilon/3$. Therefore, for every $\epsilon > 0$, there exists N such that for all $n \ge N$, $|J^n_\infty(x) - J^\infty_\infty(x)| \le \epsilon$.

Thus, we have

$$\lim_{n\to\infty} \lim_{m\to\infty} J^n_m(x) = J^*_\beta(x).$$

◇

THEOREM 3.14. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.2 holds. Then, DRO (3.5), value convergence, and RSP (3.4) robustness hold within the ambiguity classes $\mathcal{D}''^n$.*

Proof. What is left to show is that value convergence and RSP robustness hold.

Let $\mathcal{R}_n \in \mathcal{D}''^n$ be a sequence of weakly continuous kernels. It suffices to show that this sequence of kernels converges continuously weakly to $\mathcal{T}$. That is, suppose $(x_n, u_n) \to (x, u)$. The uniform ambiguity constraint gives

$$\sup_{(x,u)\in\mathbb{X}\times\mathbb{U}} \rho_{\mathcal{F}}(\mathcal{R}_n(\cdot|x,u), \mathcal{T}(\cdot|x,u)) \le \frac{1}{n},$$

Thus, we have that

$$\rho_{\mathcal{F}}(\mathcal{R}_n(\cdot|x_n,u_n),\mathcal{T}(\cdot|x,u)) \le \rho_{\mathcal{F}}(\mathcal{R}_n(\cdot|x_n,u_n),\mathcal{T}(\cdot|x_n,u_n)) + \rho_{\mathcal{F}}(\mathcal{T}(\cdot|x_n,u_n),\mathcal{T}(\cdot|x,u))$$
$$\le \frac{1}{n} + \rho_{\mathcal{F}}(\mathcal{T}(\cdot|x_n,u_n),\mathcal{T}(\cdot|x,u)) \tag{3.17}$$

where the last term goes to 0 by the weak continuity of $\mathcal{T}$. By Theorem 3.7, value convergence holds, and by Theorem 3.8, RSP holds. ◇

REMARK 3.1. *It appears that the truly important distinction between RSP and DRO in the above context is the compactness condition of the space of allowable kernels (in the uncertainty set) given in (3.11), which requires then an Arzela-Ascoli type equi-continuity condition. This is not required for the RSP setting*

3.3. Beyond weak convergence: Continuous total variation convergence in actions. For the following section, we will require the ambiguity sets to be defined under the total variation metric. More specifically,

$$\mathcal{D}''^n_x := \Big\{ \mathcal{R}(\cdot|x,\cdot) \in C(\mathbb{U},\mathcal{P}(\mathbb{X})) : \sup_{u\in\mathbb{U}} ||\mathcal{R}(\cdot|x,u),\mathcal{T}(\cdot|x,u)||_{TV} \le \frac{1}{n},$$
$$u \mapsto \mathcal{R}(\cdot|x,u) \text{ is uniformly equicontinuous} \Big\}. \tag{3.18}$$

Furthermore, suppose that $\mathcal{P}(\mathbb{X})$ is endowed with total variation convergence and

$$\bigcup_{x\in\mathbb{X}} \bigcup_{\mathcal{R}(\cdot|x,\cdot)\in\mathcal{D}''^n_x} \bigcup_{u\in\mathbb{U}} \mathcal{R}(\cdot|x,u)$$

has compact closure under total variation convergence. We define the global ambiguity sets as

$$\mathcal{D}''^n := \left\{ \mathcal{R} : x \mapsto \mathcal{R}(\cdot|x,\cdot) \text{ is measurable}, \quad \mathcal{R}(\cdot|x,\cdot) \in \mathcal{D}''^n_x \text{ for every } x \in \mathbb{X} \right\}. \tag{3.19}$$

As the total variation metric falls under our class of convex metrics (3.13), the ambiguity sets defined above are indeed convex.

ASSUMPTION 3.3. *Suppose that $\mathbb{U}$ is compact, $\mathcal{T}$ is continuous in total variation, and, for every $x \in \mathbb{X}$, $u \mapsto \mathcal{T}(\cdot|x,u)$ is total variation continuous in u. Let the uncertainty sets be characterized by $\mathcal{R} \in \mathcal{D}''^n$ as defined in (3.19), and that $x \mapsto \mathcal{D}''^n_x$ has a measurable graph. Suppose further that $c : \mathbb{X}\times\mathbb{U} \to \mathbb{R}$ is bounded and measurable and, for every $x \in \mathbb{X}$, $c(x,\cdot)$ is continuous.*

PROPOSITION 3.15. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.3 holds. Then, the iterations defined by (3.14) are well-defined and define a contraction with contraction rate β. Therefore, the iterations converge to a fixed point $J^n_\infty \in B_b(\mathbb{X})$ and satisfy*

$$\lim_{m\to\infty} ||J^n_m - J^n_\infty||_\infty = 0.$$

Proof. In our case, we can establish for our function $g_J : \mathbb{X}\times\mathcal{P}(\mathbb{U})\times\mathcal{D}''^n_x \ni (x,\gamma,q) \mapsto g_J(x,\gamma,q) \in \mathbb{R}$, measurable in x, continuous in γ and q, $x \mapsto \mathcal{D}''^n_x$ has a measurable graph, and $\mathcal{P}(\mathbb{U})$ and $\mathcal{D}''^n_x$ are both compact, $G(x) := \min_{\gamma\in\mathcal{P}(\mathbb{U})} \max_{q\in\mathcal{D}''^n_x} g_J(x,\gamma,q)$ is measurable; see [35, Theorem 2], [58].

The proof differs from that of the weak convergence case in one important respect: since the cost function is not assumed to be continuous in x, the iterations need not preserve continuity in x. That is, the appropriate space is $B_b(\mathbb{X})$, the space of bounded measurable real-valued functions on $\mathbb{X}$, equipped with the supremum norm.

Note, for $J \in B_b(\mathbb{X})$, we have

$$\left| \int J d\pi_1 - \int J \pi_2 \right| = \left| \int J(d\pi_1 - d\pi_2) \right| \leq ||J||_\infty ||\pi_1 - \pi_2||_{TV}.$$

Therefore, we have that $\pi \mapsto \int J d\pi$ is continuous under total variation convergence. Furthermore, since $q \in \mathcal{D}''^n_x$ is continuous in total variation in u and c is continuous in u, we have that $u \mapsto c(x,u) + \beta \int J(x')q(x'|u)$ is continuous. Thus, g_J (3.15) is continuous in (γ, q) for each $x \in \mathbb{X}$. By the compactness of $\mathcal{P}(\mathbb{U})$ and $\mathcal{D}''^n_x$, and the measurable graph of $x \mapsto \mathcal{D}''^n_x$, we have that for $J^n_m \in B_b(\mathbb{X})$,

$$J^n_{m+1}(x) = \min_{\gamma(\cdot|x) \in \mathcal{P}(\mathbb{U})} \sup_{q \in \mathcal{D}''^n_x} g_{J^n_m}(x, \gamma, q) \in B_b(\mathbb{X}).$$

The contraction argument is identical to that in Proposition 3.10. Therefore, since $||J^n_{m+1} - \bar{J}^n_{m+1}||_\infty \leq \beta ||J^n_m - \bar{J}^n_m||_\infty$, we have that the iterations converge to a fixed point $J^n_\infty \in B_b(\mathbb{X})$ and satisfy

$$\lim_{m \to \infty} ||J^n_m - J^n_\infty||_\infty = 0$$

◇

PROPOSITION 3.16. *Suppose Assumption 3.3 holds. Then, for $J \in B_b(\mathbb{X})$,*

$$\min_{\gamma(\cdot|x) \in \mathcal{P}(\mathbb{U})} \sup_{q \in \mathcal{D}''^n_x} g_J(x, \gamma, q) = \sup_{q \in \mathcal{D}''^n_x} \min_{\gamma(\cdot|x) \in \mathcal{P}(\mathbb{U})} g_J(x, \gamma, q)$$

Proof. As g_J (3.15) is affine in γ and q, and since $\mathcal{P}(\mathbb{U})$ and $\mathcal{D}''^n_x$ are compact and convex, Sion's minimax theorem gives the result. ◇

THEOREM 3.17. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.3 holds. Then, $J^n_\infty(x)$ is indeed the DRO value for $\mathcal{D}''^n$. That is,*

$$J^n_\infty(x) = \inf_{\gamma \in \Gamma} \sup_{\mathcal{R} \in \mathcal{D}''^n} J_\beta(x, \mathcal{R}, c, \gamma).$$

Proof. Now, for every x, Sion's theorem and compactness gives a saddle-point $(\gamma^*_n(\cdot|x), q^*_n(x)) \in \mathcal{P}(\mathbb{U}) \times \mathcal{D}''^n_x$ for $g_{J^n_\infty}(x, \cdot, \cdot)$. That is,

$$g_{J^n_\infty}(x, \gamma^*_n(\cdot|x), q) \leq J^n_\infty(x) \leq g_{J^n_\infty}(x, \gamma, q^*_n(x))$$

where $g_{J^n_\infty}(x, \gamma^*_n(\cdot|x), q^*_n(x)) = J^n_\infty(x)$. A measurable selection theorem (see [54, Theorem 4.1]) yields measurable selectors for $x \mapsto \gamma^*_n(\cdot|x)$ and $x \mapsto q^*_n(x)$. We define the glued kernel $\mathcal{R}^*_n(\cdot|x,u) := q^*_n(x)(\cdot|u)$. $\mathcal{R}^*_n$ is indeed a measurable stochastic kernel, and thus $\mathcal{R}^*_n \in \mathcal{D}''^n$.

By the saddle-point inequalities, we can once again establish that $J^n_\infty(x)$ is indeed the DRO value for $\mathcal{D}''^n$. ◇

THEOREM 3.18. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.3 holds. The DRO solution can be obtained via the iterations given by (3.14) so that*

$$\lim_{n \to \infty} \lim_{m \to \infty} J^n_m(x) = J^*_\beta(x).$$

Proof. The argument is identical to that in Theorem 3.13.

THEOREM 3.19. *Consider (3.3) and the DRO formulation (3.5). Suppose that Assumption 3.3 holds. Then, DRO (3.5), value convergence, and RSP (3.4) robustness hold within the ambiguity classes $\mathcal{D}''^n$.*

Proof. What is left to show is that value convergence and RSP robustness hold.

Let $\mathcal{R}_n \in \mathcal{D}''^n$, and let $\gamma^*_{\mathcal{R}_n}$ be an optimal policy for $\mathcal{R}_n$. The uniform ambiguity constraint gives

$$\sup_{(x,u)\in\mathbb{X}\times\mathbb{U}} \|\mathcal{R}_n(\cdot|x,u) - \mathcal{T}(\cdot|x,u)\|_{TV} \leq \frac{1}{n}.$$

By [41, Theorem 2.6], we have

$$\left|J_\beta(x,\mathcal{R}_n,\gamma^*_{\mathcal{R}_n}) - J^*_\beta(x)\right| \leq \frac{||c||_\infty \beta}{(\beta-1)^2} \sup_{(x,u)\in\mathbb{X}\times\mathbb{U}} ||\mathcal{R}_n(\cdot|x,u) - \mathcal{T}(\cdot|x,u)||_{TV}.$$

Respectively, by [39, Theorem 2.5.1], we have

$$\left|J_\beta(x,\mathcal{T},\gamma^*_{\mathcal{R}_n}) - J^*_\beta(x)\right| \leq \frac{2\|c\|_\infty \beta}{(1-\beta)^2} \sup_{(x,u)\in\mathbb{X}\times\mathbb{U}} \|\mathcal{R}_n(\cdot|x,u) - \mathcal{T}(\cdot|x,u)\|_{TV}\,.$$

Thus,

$$\left|J_\beta(x,\mathcal{R}_n,\gamma^*_{\mathcal{R}_n}) - J^*_\beta(x)\right| \to 0, \qquad \text{and} \qquad \left|J_\beta(x,\mathcal{T},c,\gamma^*_{\mathcal{R}_n}) - J^*_\beta(x)\right| \to 0.$$

That is, both value convergence and RSP robustness hold. ◇

4. Reflection and Conclusion. In our paper we revisited the mathematically interesting and practically highly relevant problem of robustness under two commonly studied perspectives and formulations. Notably, we showed that these two formulations, DRO vs RSP as we referred to in the paper, are not asymptotically equivalent, and we established relationship properties including conditions under which these formulations do imply one another. During our analysis, some further observations have been made:

- DRO provides a minimax approach, provides uniform guarantees across a variety of models, and is thus a natural general formulation.
- Since DRO is based on a game theoretic formulation, a perhaps natural conjecture that asymptotic DRO robustness would in general imply RSP robustness is not correct.
- On the other hand, DRO may be a demanding concept:
  - DRO demands a metric space structure on the space of models, which RSP does not (such as setwise convergence being allowed under RSP).
  - The dynamic formulation of DRO can be ambigious as various and somewhat inconsistent formulations (e.g. with regard to the leader follower formulation which is global versus stage-wise) in the literature highlight.
  - Furthermore, the well-posedness of a dynamic programming based formulation of DRO requires more stringent conditions leading to measurable selection and regularity, when compared with RSP robustness.
- On the other hand, RSP is much more common in applied mathematics and general natural and applied sciences: One could argue that model based thinking and technical design perhaps implicitly assumes RSP robustness.
- As our paper has shown, however, the particular convergence to an ideal model may be critical as it may affect the regularity of the optimized closed-loop path of approximate models together with their controls, the regularity of which impacts the overall value convergence and RSP robustness properties.

Future work entails studying and arriving at counterparts of the analysis in the paper for continuous-time models as well as for problems under partial and decentralized information.

[1] D. J. Aldous. *Weak convergence and the general theory of processes*. Editeur inconnu, 1981.

[2] A. Almudevar. and E. F. Arruda. Optimal approximation schedules for a class of iterative algorithms, with an application to multigrid value iteration. *IEEE Transactions on Automatic Control*, 57:3132–3146, 2012.

[3] E. F. Arruda, F. Ourique, J. Lacombe, and A. Almudevar. Accelerating the convergence of value iteration by using partial transition functions. *European Journal of Operational Research*, 229:190–198, 2013.

[4] R. J. Aumann. Mixed and behavior strategies in infinite extensive games. Technical report, Princeton University NJ, 1961.

[5] J. Backhoff-Veraguas, D. Bartl, M. Beiglböck, and M. Eder. Adapted Wasserstein distances and stability in mathematical finance. *Finance and Stochastics*, 24(3):601–632, 2020.

[6] J. Backhoff-Veraguas, D. Bartl, M. Beiglböck, and M. Eder. All adapted topologies are equal. *Probability Theory and Related Fields*, 178(3):1125–1172, 2020.

[7] T. Başar. Stochastic stagewise Stackelberg strategies for linear quadratic systems. In *Stochastic Control Theory and Stochastic Differential Systems, M. Kohlmann and W. Vogel, editors*, pages 264–276. Springer, 1979.

[8] T. Başar and P. Bernhard. *H-infinity Optimal Control and Related Minimax Design Problems: A Dynamic Game Approach*. Birkhäuser, Boston, MA, 1995.

[9] T. Başar and G.J. Olsder. *Dynamic Noncooperative Game Theory*. SIAM Classics in Applied Mathematics, Philadelphia, PA, 1999.

[10] D. Bartl, M. Beiglböck, and G. Pammer. The wasserstein space of stochastic processes. *Journal of the European Mathematical Society*, 2024.

[11] Daniel Bartl and Johannes Wiesel. Sensitivity of multiperiod optimization problems with respect to the adapted wasserstein distance. *SIAM Journal on Financial Mathematics*, 14(2):704–720, 2023.

[12] T. Başar. Robust designs through risk sensitivity: An overview. *Journal of Systems Science and Complexity*, 34(5):1634–1665, 2021.

[13] N. Bäuerle and A. Glauner. Distributionally robust markov decision processes and their connection to risk measures. *Mathematics of Operations Research*, 47(3):1757–1780, 2022.

[14] E. Bayraktar, Yan Y. Dolinsky, and J. Guo. Continuity of utility maximization under weak convergence. *Mathematics and Financial Economics*, pages 1–33, 2020.

[15] M. Beiglböck, B. Jourdain, W. Margheriti, and G. Pammer. Approximation of martingale couplings on the line in the adapted weak topology. *Probability Theory and Related Fields*, 183(1):359–413, 2022.

[16] A. Benavoli and L. Chisci. Robust stochastic control based on imprecise probabilities. *IFAC Proceedings Volumes*, 44(1):4606–4613, 2011.

[17] J. Blanchet, J. Li, S. Lin, and X. Zhang. Distributionally robust optimization and robust statistics. *Statistical Science*, 40(3):351–377, 2025.

[18] J. Blanchet and K. Murthy. Quantifying distributional model risk via optimal transport. *SSRN Electronic Journal*, 04 2016.

[19] R. K. Boel, M. R. James, and I. R. Petersen. Robustness and risk-sensitive filtering. *IEEE Transactions on Automatic Control*, 47(3):451–461, 2002.

[20] V. S. Borkar. White-noise representations in stochastic realization theory. *SIAM J. on Control and Optimization*, 31:1093–1102, 1993.

[21] B. Bozkurt, A. Mahajan, A. Nayyar, and Y. Ouyang. Model approximation in mdps with unbounded per-step cost. *IEEE Transactions on Automatic Control*, 2025.

[22] L. Devroye and L. Györfi. *Non-parametric Density Estimation: The L_1 View*. John Wiley, New York, 1985.

[23] R. M. Dudley. The speed of mean Glivenko-Cantelli convergence. *The Annals of Mathematical Statistics*, 40(1):40–50, 1969.

[24] R. M. Dudley. *Real Analysis and Probability*. Cambridge University Press, Cambridge, 2nd edition, 2002.

[25] P. Dupuis, M. R. James, and I. Petersen. Robust properties of risk-sensitive control. *Mathematics of Control, Signals and Systems*, 13(4):318–332, 2000.

[26] E. Erdoğan and G. N. Iyengar. Ambiguous chance constrained problems and robust optimization. *Mathematical Programming*, 107(1-2):37–61, 2005.

[27] P. M. Esfahani and D. Kuhn. Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, pages 1–52, 2017.

[28] S. N. Ethier and T. G. Kurtz. *Markov processes: characterization and convergence*, volume 282. John Wiley & Sons, 2009.

[29] Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3–4):707–738, October 2014.

[30] I. I. Gihman and A. V. Skorohod. *Controlled stochastic processes*. Springer Science & Business Media, 2012.

[31] O. Gossner and T. Tomala. Entropy bounds on Bayesian learning. *Journal of Mathematical Economics*, 44(1):24–32, 2008.

[32] L. P. Hansen and T. J. Sargent. Robust control and model uncertainty. *American Economic Review*, 91(2):60–66, 2001.

[33] M. F. Hellwig. Sequential decisions under uncertainty and the maximum theorem. *Journal of Mathematical Economics*, 25(4):443–464, 1996.

[34] O. Hernandez-Lerma and J. Lasserre. *Discrete-time Markov control processes*. Springer, 1996.

[35] C. J. Himmelberg, T. Parthasarathy, and F. S. VanVleck. Optimal plans for dynamic programming problems. *Mathematics of Operations Research*, 1(4):390–394, 1976.

[36] G. N. Iyengar. Robust dynamic programming. *Mathematics of Operations Research*, 30(2):257–280, 2005.
[37] D. Jacobson. Optimal stochastic linear systems with exponential performance criteria and their relation to deterministic differential games. *IEEE Transactions on Automatic control*, 18(2):124–131, 1973.
[38] A. D. Kara and S. Yüksel. Robustness to incorrect priors in partially observed stochastic control. *SIAM Journal on Control and Optimization*, 57(3):1929–1964, 2019.
[39] A. D. Kara and S. Yüksel. Robustness to approximations and model learning in MDPs and POMDPs. In A. B. Piunovskiy and Y. Zhang, editors, *Modern Trends in Controlled Stochastic Processes: Theory and Applications, Volume III*. Luniver Press, 2021.
[40] A.D Kara and S. Yüksel. Robustness to incorrect priors in partially observed stochastic control. *SIAM Journal on Control and Optimization*, 57(3):1929–1964, 2019.
[41] A.D Kara and S. Yüksel. Robustness to incorrect system models in stochastic control. *SIAM Journal on Control and Optimization*, 58(2):1144–1182, 2020.
[42] H. Lam. Robust sensitivity analysis for stochastic systems. *Mathematics of Operations Research*, 41(4):1248–1275, 2016.
[43] H.J. Langen. Convergence of dynamic programming models. *Mathematics of Operations Research*, 6(4):493–512, 1981.
[44] Y. Li and A. Shapiro. Rectangularity and duality of distributionally robust markov decision processes: Y. li, a. shapiro. *Mathematical Programming*, pages 1–42, 2025.
[45] Omar Mrani-Zentar and Serdar Yüksel. Kernel metrics and learning for borel mdps: Identifiability and adaptive control. *arXiv preprint arXiv:2503.11058*, 2025.
[46] A. Müller. How does the value function of a markov decision process depend on the transition probabilities? *Mathematics of Operations Research*, 22(4):872–885, 1997.
[47] A. Nilim and L. El Ghaoui. Robust control of Markov decision processes with uncertain transition matrices. *Operations Research*, 53(5):780–798, 2005.
[48] B. Øksendal and A. Sulem. Forward–backward stochastic differential games and stochastic control under model uncertainty. *Journal of Optimization Theory and Applications*, 161(1):22–55, 2014.
[49] G. Pammer. A note on the adapted weak topology in discrete time. *Electronic Communications in Probability*, 29:1–13, 2024.
[50] Nikolaos S. Papageorgiou and Patrick Winkert. *Applied Nonlinear Functional Analysis: An Introduction*. De Gruyter, 2018.
[51] I. Petersen, M. R. James, and P. Dupuis. Minimax optimal control of stochastic uncertain systems with relative entropy constraints. *IEEE Transactions on Automatic Control*, 45(3):398–412, 2000.
[52] P. Dai Pra, L. Meneghini, and W. J. Runggaldier. Connections between stochastic control and dynamic games. *Mathematics of Control, Signals and Systems*, 9(4):303–326, 1996.
[53] S. Pradhan and S. Yüksel. Continuity of cost in Borkar control topology and implications on discrete space and time approximations for controlled diffusions under several criteria. *Electronic Journal of Probability*, 29:1–32, 2024.
[54] U. Rieder. Equilibrium plans for non-zero-sum markov games. *Game theory and related topics*, pages 91–101, 1979.
[55] N. Saldi, T. Linder, and S. Yuk̈sel. *Finite Approximations in Discrete-time Stochastic Control: Quantized Models and Asymptotic Optimality.* Birkhäuser, 2018.
[56] N. Saldi and S. Yüksel. Kernel mean embedding topology: Weak and strong forms for stochastic kernels and implications for model learning. *Annals of Applied Probability (also arXiv:2502.13486)*, 2026.
[57] N. Saldi, S. Yüksel, and T. Linder. Asymptotic optimality of finite approximations to Markov decision processes with Borel spaces. *Math. Oper. Res.*, pages 1–34, March 2017.
[58] M. Schäl. Conditions for optimality in dynamic programming and for the limit of n-stage optimal policies to be optimal. *Z. Wahrscheinlichkeitsth*, 32:179–296, 1975.
[59] H. Sun and H. Xu. Convergence analysis for distributionally robust optimization and equilibrium problems. *Mathematics of Operations Research*, 41:377–401, 07 2015.
[60] I. Tzortzis, C.D. Charalambous, and T. Charalambous. Dynamic programming subject to total variation distance ambiguity. *SIAM Journal on Control and Optimization*, 53(4):2040–2075, 2015.
[61] V. N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, New York, 2nd ed., 2000.
[62] Jonathan Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A), November 2019.
[63] W. Wiesemann, D. Kuhn, and B. Rustem. Robust Markov decision processes. *Mathematics of Operations Research*, 38(1):153–183, 2012.
[64] Y. Wu and S. Verdú. Functional properties of minimum mean-square error and mutual information. *IEEE Transactions on Information Theory*, 58(3):1289–1301, 2012.
[65] H. Xu and S. Mannor. Distributionally robust Markov decision processes. In J. D. Lafferty, C. K. I. Williams, J. Shawe-Taylor, R. S. Zemel, and A. Culotta, editors, *Advances in Neural Information Processing Systems 23*, pages 2505–2513. Curran Associates, Inc., 2010.
[66] S. Yüksel. On Borkar and Young relaxed control topologies and continuous dependence of invariant measures on control policy. *SIAM Journal on Control and Optimization*, 62(4):2367–2386, 2024.
[67] S. Yüksel. *Optimization and Control of Stochastic Systems*. Queen's University, Lecture Notes, available online, 2026.
[68] S. Yüksel and T. Başar. *Stochastic Teams, Games, and Control under Information Constraints*. Springer,

Cham, 2024.
[69] S. Yüksel and T. Linder. Optimization and convergence of observation channels in stochastic control. *SIAM J. on Control and Optimization*, 50:864–887, 2012.
[70] S. Yüksel and T. Linder. Optimization and convergence of observation channels in stochastic control. *SIAM J. on Control and Optimization*, 50:864–887, 2012.
[71] K. Zhou, J. C. Doyle, and K. Glover. *Robust and optimal control*, volume 40. Prentice-Hall, 1996.
[72] Y. Zhou, Y. Song, and S. Yüksel. Robustness to model approximation, learning, and sample complexity in wasserstein regular mdps. *arXiv preprint arXiv:2410.14116*, 2024.